

MIT 8301 Virtual Laboratory

German Credit Risk Classification

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Programme	Master of Information Technology
Specialisation	Artificial Intelligence & Machine Learning
Course	MIT 8301 - Machine Learning and AI Fundamentals
Academic Session	2026/2027

Cover Page and Student Details

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Executive Summary

This assessment develops a reproducible classifier for bad credit risk. The supplied file contained 1,000 applicants and nine predictors but omitted the label. An authentic target-inclusive edition was reconciled with a 100% exact row-level predictor match and zero mismatched cells before labels were appended. The verified target contains 700 good and 300 bad applicants. Four models were trained and evaluated with leakage-safe preprocessing. Tuned Logistic Regression was selected under a predeclared performance-and-governance rule.

Problem Statement and Objectives

The objective is to identify risky applicants ($\text{credit_risk}=1$) while recognising the asymmetric institutional and fairness costs of false negatives and false positives. The work covers EDA, missing values, outliers, encoding, scaling, scratch Logistic Regression, tuned model comparison, transparent feature interpretation, and responsible-use limitations.

Dataset Description, Provenance, and Target Restoration

The supplied export had 1,000 rows, an index column, and nine predictors. The target source was the Hugging Face mirror of Kaggle's German Credit Risk - With Target edition. After removing the export index and normalising only whitespace/case, every predictor cell matched. No fuzzy or positional merge was used. good was encoded 0 and bad 1. SHA-256 checksums and per-feature match percentages are recorded in the provenance report.

Exploratory Data Analysis

There are no duplicated complete rows. Age has 53 unique values, credit amount 921, and duration 33. The target is moderately imbalanced (70% good, 30% bad). Histograms, robust-scaled boxplots, categorical distributions, a missingness chart, a correlation heatmap, target comparisons, and observed risk by purpose appear in the figures appendix. These are descriptive associations, not causal findings.

Missing Values and Outliers

Savings status has 183 missing values; checking status has 394. A dedicated Unknown_or_No_Account category preserves potentially informative absence and is learned inside training folds. IQR review found {'Age': 23, 'Credit amount': 72, 'Duration': 70}. Values are plausible, so they are not deleted. Training-fitted 1.5-IQR clipping limits leverage, followed by robust scaling. This avoids data leakage.

Encoding, Scaling, and Split

The data were split into 800 training and 200 untouched test rows with stratification and random state 42. Age, amount, and duration use median imputation, IQR clipping, and RobustScaler. Nominal fields, including Job, use constant imputation and one-hot encoding with unknown-category tolerance. All transformations are fitted only on training data or within cross-validation folds.

Training, Validation and Test Strategy

The dataset was partitioned into an 80% training set (800 observations) and a 20% independent test set (200 observations) using a stratified train-test split with `random_state=42`. The training set contains 560 good and 240 bad applicants, and the test set contains 140 good and 60 bad applicants; both therefore preserve the original 70% good / 30% bad class distribution.

Hyperparameter optimisation was performed exclusively on the training data using Stratified 5-Fold Cross-Validation within GridSearchCV. During this process, the 800 training observations were repeatedly divided into internal training and validation folds while preserving class proportions. Every training observation serves in validation across the rotating folds. After optimal hyperparameters were identified, GridSearchCV automatically refitted each model on the complete 800-row training set before a single evaluation on the untouched 200-row test set.

No separate validation dataset was required because cross-validation already creates internal validation folds. Although 60/20/20 and 70/15/15 partitions are also established approaches, this project intentionally adopted 80% train + 20% test + stratified 5-fold cross-validation because the dataset contains only 1,000 observations, cross-validation provides statistically stronger hyperparameter evaluation, and more observations remain available for model learning. This aligns with current scikit-learn and production machine-learning practice.

Missing-value imputation, categorical encoding, numerical scaling, IQR outlier clipping, and all feature transformations are implemented inside scikit-learn Pipeline and ColumnTransformer objects. They are fitted within training folds and subsequently applied to validation or test rows. GridSearchCV received only `X_train` and `y_train`; the decision threshold remained fixed at 0.5, so the test set influenced neither model selection, hyperparameter tuning, preprocessing, nor threshold optimisation.

Logistic Regression from Scratch

The implementation provides a stable clipped sigmoid, explicit bias, binary cross-entropy, vectorised gradients, L2 regularisation, convergence tolerance, loss history, probability estimates, and input/fitted-state validation. It converged=True in 4233 iterations with final loss 0.50469. Its test performance is compared directly with scikit-learn.

Model Training and Hyperparameter Tuning

Training used stratified five-fold GridSearchCV and ROC-AUC as the primary score. The test set was never used for tuning. Tuning controls regularisation, margin/kernel complexity, class weighting, and Naive Bayes variance smoothing, improving generalisation without test leakage.

Tuned Logistic Regression

Best cross-validated ROC-AUC: 0.756

Parameter	Selected value
C	0.1
Class weight	None
Penalty	l2
Solver	liblinear

Tuned SVM

Best cross-validated ROC-AUC: 0.765

Parameter	Selected value
C	1
Class weight	balanced
Gamma	0.1
Kernel	rbf

Tuned Gaussian Naive Bayes

Best cross-validated ROC-AUC: 0.666

Parameter	Selected value
Variance smoothing	1e-12

Test Evaluation and Comparison

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Tuned SVM	0.710	0.511	0.750	0.608	0.786
Tuned Logistic Regression	0.730	0.594	0.317	0.413	0.766
Logistic Regression from Scratch	0.720	0.548	0.383	0.451	0.756
Tuned Gaussian Naive Bayes	0.675	0.472	0.700	0.564	0.702

A false negative classifies a risky applicant as good and may expose the institution to loss. A false positive classifies a good applicant as risky and may cause unfair rejection or harsher terms. Accuracy alone is therefore insufficient; recall, F1, ROC-AUC, thresholds, interpretability, and operating costs matter.

Best-Model Justification

The predeclared rule prefers tuned Logistic Regression when its ROC-AUC is within 0.03 of the highest model, otherwise the highest-AUC model is selected. The result is Tuned Logistic Regression. This accepts a small discrimination trade-off for transparent coefficients, simpler deployment, auditability, and regulatory suitability. Operational threshold tuning could improve risky-class recall and must be set from validated costs, not the test set.

Feature Importance and Interpretation

The largest absolute transformed Logistic Regression coefficients are:

- cat__Checking account_Unknown_or_No_Account: -0.879
- num__Duration: +0.514
- cat__Checking account_little: +0.471
- cat__Saving accounts_Unknown_or_No_Account: -0.392
- cat__Purpose_radio/tv: -0.360
- cat__Housing_own: -0.343
- cat__Sex_male: -0.330
- cat__Saving accounts_little: +0.313

- cat__Purpose_education: +0.307
- cat__Saving accounts_rich: -0.283

Positive values increase estimated bad-credit log-odds and negative values decrease them, conditional on the encoding and scaling. Coefficients express association, not causation. One-hot terms are interpreted relative to the omitted all-zero representation.

Ethics, Fairness, and Responsible Use

Sex is sensitive and age can create protected-class concerns. Historical labels may encode past discrimination and other fields may serve as proxies. Supplementary metrics by sex are reported but small test subgroups cannot establish fairness. The model requires human review, reasons for adverse decisions, appeal routes, privacy controls, drift monitoring, periodic bias audits, and independent validation. Prediction is not causal judgement.

Limitations

The dataset is small, historical, geographically specific, and omits important affordability and contemporary lending variables. Labels have no event-time detail; calibration and temporal stability are not established. The simplified nine-feature representation loses information from the original 20-feature UCI data. Reported test estimates have sampling uncertainty.

Lessons Learned and Engineering Reflection

Principal engineering challenge

The most consequential challenge was not model selection but the absence of a target variable in the supplied German Credit CSV. Supervised classification requires authentic observed outcomes: without a good/bad credit label, there is no valid response variable from which a classifier can learn or against which it can be evaluated. Creating synthetic labels, inferring them from predictor thresholds, clustering applicants and treating the clusters as outcomes, or deriving labels from the same features used for prediction would have introduced circular reasoning and invalidated the experiment. The appropriate engineering decision was therefore to pause supervised modelling until label provenance could be established. Preserving dataset integrity was more important than forcing a pipeline to produce metrics.

Engineering solution and provenance control

The resolution began with an investigation of the supplied file's schema and the provenance of the simplified German Credit representation. An authentic target-inclusive edition with the same nine predictors was identified. The reconciliation process removed export-only identifiers, standardised column names, and normalised only harmless representation differences such as surrounding whitespace and demonstrably equivalent category case. It then compared every predictor, row by row and feature by feature, before accepting any label.

The verification produced a 100% match for every predictor and zero mismatched cells across 1,000 observations. SHA-256 checksums were recorded for both raw inputs, duplicate diagnostics and class counts were produced, and the procedure was documented in a machine-readable and human-readable validation report. Only after these checks passed was the authentic target appended and mapped to `credit_risk`, with 1 representing bad/risky credit and 0 representing good credit. This retained the original alignment between each applicant and the corresponding outcome.

This experience illustrates why data provenance is an operational requirement rather than an administrative detail. In a real machine-learning system, an apparently small join, row-order change, or unverified label source can silently corrupt the relationship between features and outcomes while leaving the code executable. Checksums, explicit reconciliation rules, fail-fast assertions, and durable validation reports make those risks observable and auditable.

Machine-learning lessons

Data quality. Model quality depends first on the validity, alignment, and meaning of the data. A sophisticated estimator cannot compensate for an unauthenticated target or an incorrect feature-outcome join. The project also reinforced that reproducible preprocessing is more valuable than pursuing a marginal improvement in headline accuracy: the same inputs should pass through the same documented, testable transformations whenever the analysis is repeated.

Preventing leakage. Leakage was prevented by separating training and test data before any learned transformation entered the model workflow. Missing-value imputation, one-hot encoding, robust scaling, and IQR clipping were placed inside Pipeline and ColumnTransformer objects, causing each operation to learn its parameters from the relevant training fold only. The independent test set remained untouched until final evaluation. This matters because leakage can make a weak model appear deceptively successful by allowing information about held-out observations to influence preprocessing, tuning, or selection.

Stratification. The original target distribution contains 70% good credit and 30% bad credit. Stratified sampling preserved this balance exactly in the 800-row training set and 200-row test set. The resulting 560/240 and 140/60 class counts reduced the risk that an accidental shift in class proportions would distort comparison or make risky-class recall less reliable.

Cross-validation. Stratified 5-Fold Cross-Validation was selected instead of reserving a separate validation dataset. With only 1,000 observations, rotating validation folds make better use of the 800 training cases, provide more stable hyperparameter evidence than one small holdout, and preserve class proportions in every fold. GridSearchCV evaluated configurations only within those training folds and then refitted the selected configuration on the complete training set, supporting a stronger estimate of generalisation before the single test evaluation.

Interpretability. The tuned SVM achieved the highest test ROC-AUC, but tuned Logistic Regression was selected under the predeclared rule because its ROC-AUC was within 0.03 of the leader and its coefficients provide clearer decision logic. In credit-risk settings, engineering quality includes the ability to explain influential factors, reproduce a decision path, and support review by governance teams, auditors, and regulators. Predictive performance remains important, but a small gain from a less transparent model may not outweigh the operational value of understandable and contestable outputs.

Responsible machine-learning reflection

Responsible credit-risk modelling requires controls tied to the actual decision context. Sex and age can expose applicants to disparate outcomes, and apparently neutral variables may act as proxies for protected characteristics. Transparency therefore includes documenting the target, transformations, evaluation population, decision threshold, subgroup diagnostics, and known limitations. Explainability must be usable by reviewers and affected applicants rather than treated as a decorative chart. Reproducible code, versioned data checks, tests, model documentation, approval controls, drift monitoring, and appeal procedures form part of governance. Most importantly, a prediction should support - not replace - professional judgement in a high-impact lending decision; adverse outcomes require human oversight and a meaningful route for review.

Professional reflection

I found that this assignment strengthened my understanding of the complete machine-learning workflow. Resolving the missing target reinforced the importance of data provenance; constructing the leakage-safe preprocessing workflow clarified the separation between fitting and transformation; and implementing Logistic Regression from first principles connected the underlying mathematics to observed model behaviour. The tuning, evaluation, and responsible-use analysis also showed that valid model assessment requires disciplined validation, clear interpretation, and attention to the consequences of credit decisions.

Conclusion

The project successfully implemented the required machine-learning workflow, including verified data preparation, leakage-safe preprocessing, Logistic Regression from first principles, hyperparameter tuning,

test-set evaluation, model comparison, and feature interpretation. The resulting analysis satisfies the objectives of the Continuous Assessment while demonstrating sound machine-learning practice and appropriate caution for credit-risk decision support.

Reproducibility

The project was implemented in Python using standard data-science libraries. The notebook, source code, data-validation records, and generated outputs are reproducible from the supplied project files.

Rubric Mapping

Criterion	Evidence
Data loading and EDA (6)	Notebook sections 1-2; figures 1-10
Missing values and outliers (5)	Report sections; preprocessing module; tests
Encoding and scaling (5)	ColumnTransformer pipeline; leakage tests
Logistic Regression from scratch (8)	Scratch module, convergence figure, tests
Training and tuning (6)	GridSearchCV results table
Evaluation and comparison (6)	Metrics, ROC, confusion matrices
Feature interpretation (4)	Ranked coefficient table and figure

Code and Output Evidence Appendices

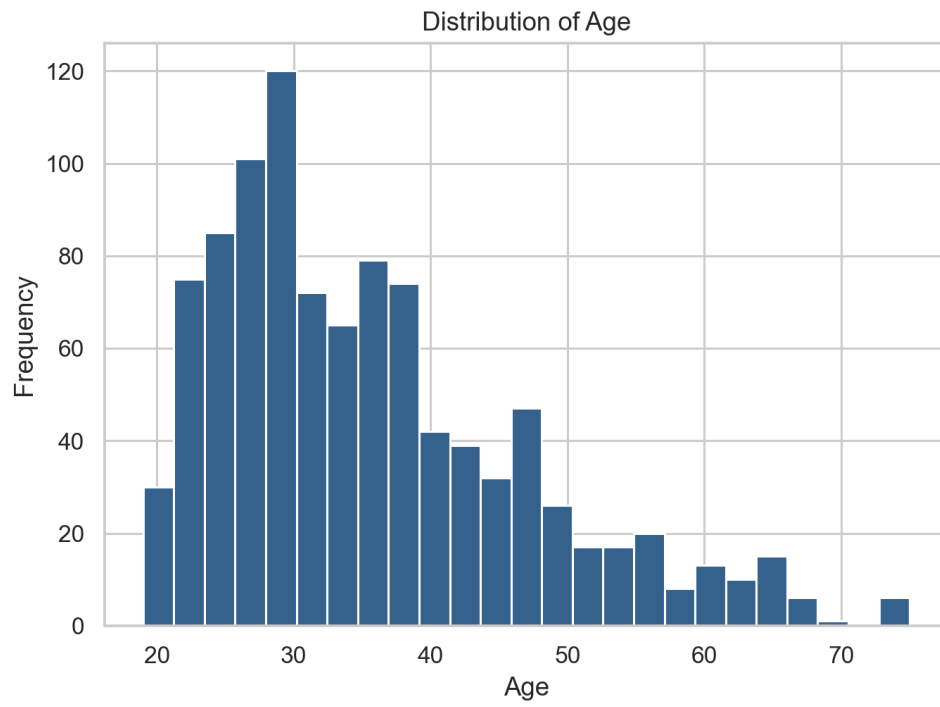
The final PDF appends the complete analysis code and curated figures and output evidence required by the assessment.

Figures and Actual Output Evidence

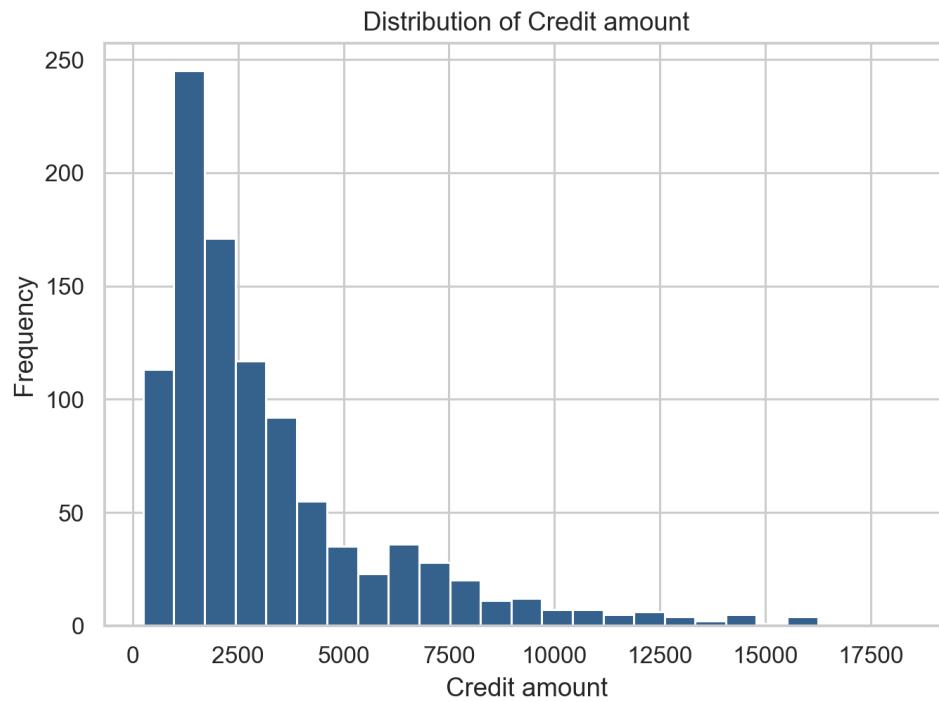
01 Target Distribution



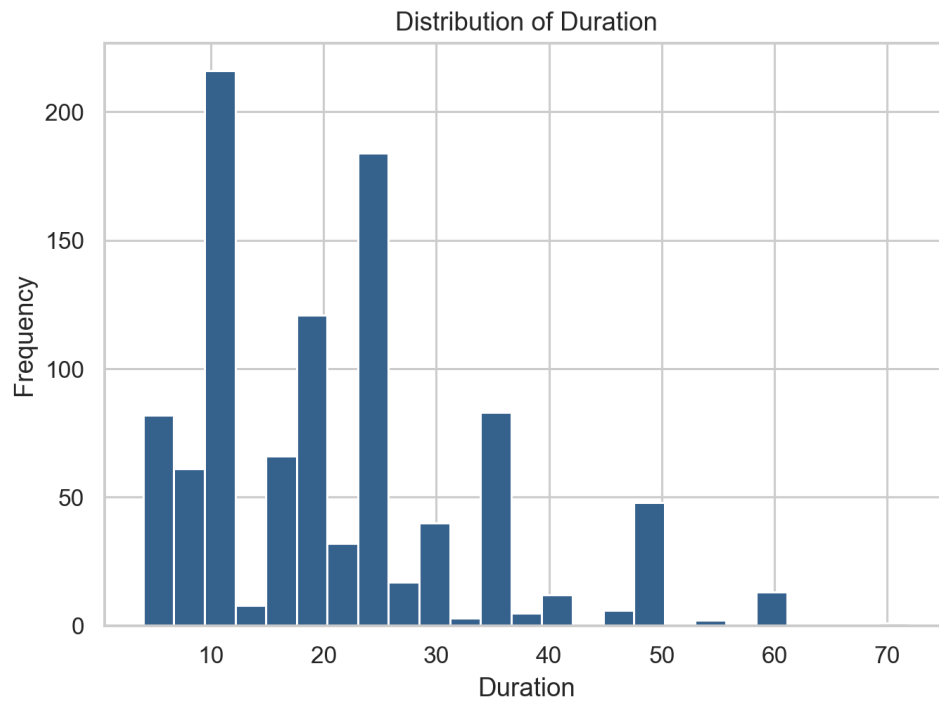
02 Age Histogram



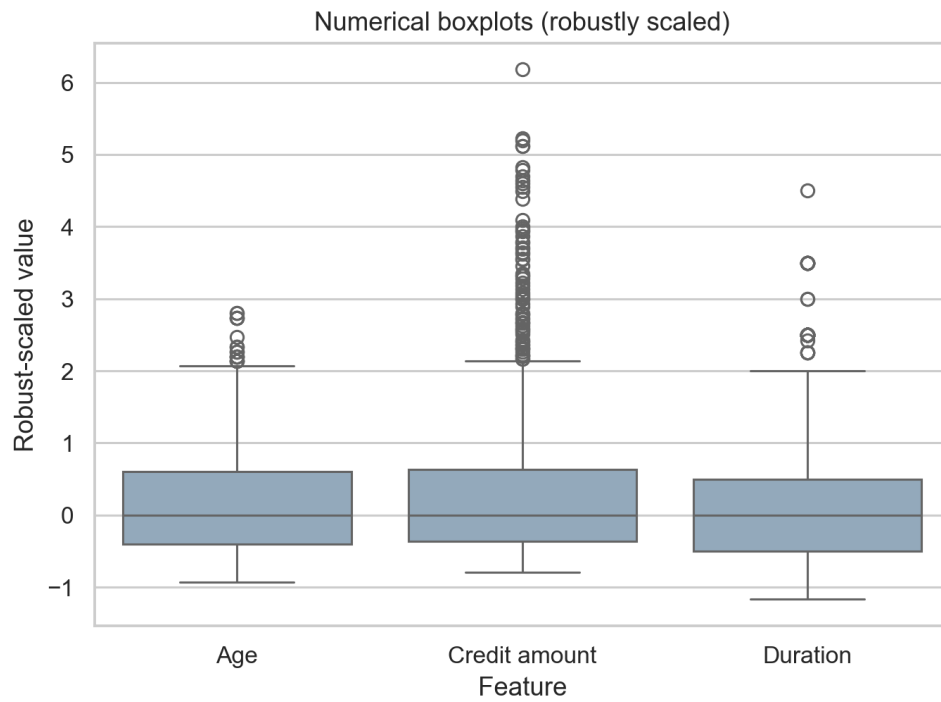
03 Credit Amount Histogram



04 Duration Histogram



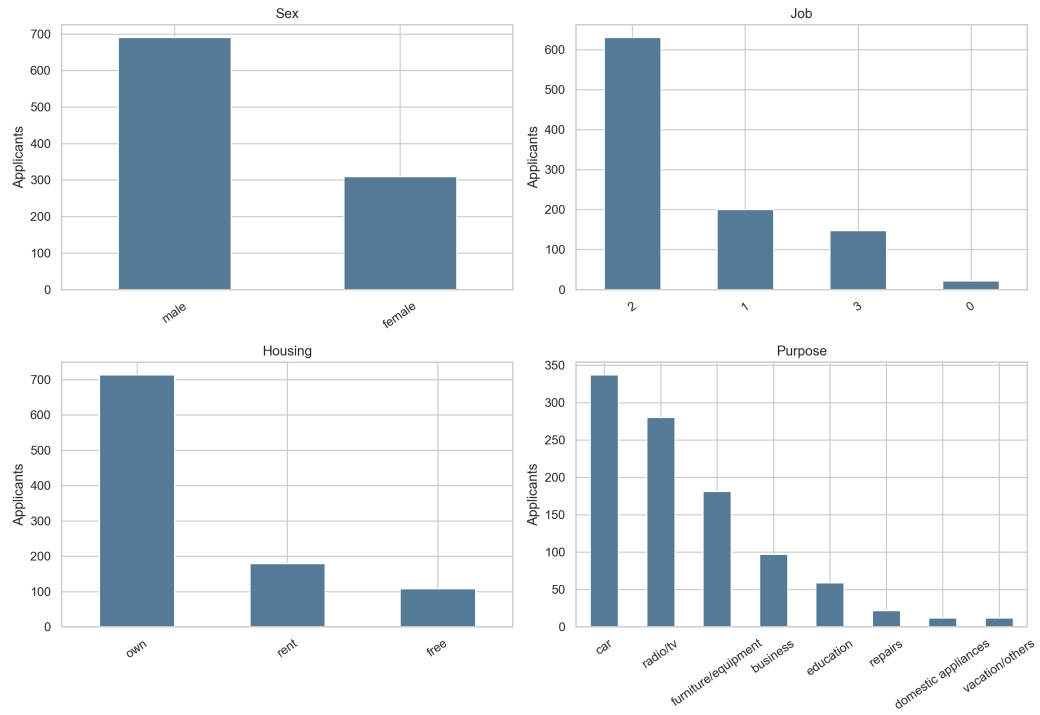
05 Numerical Boxplots



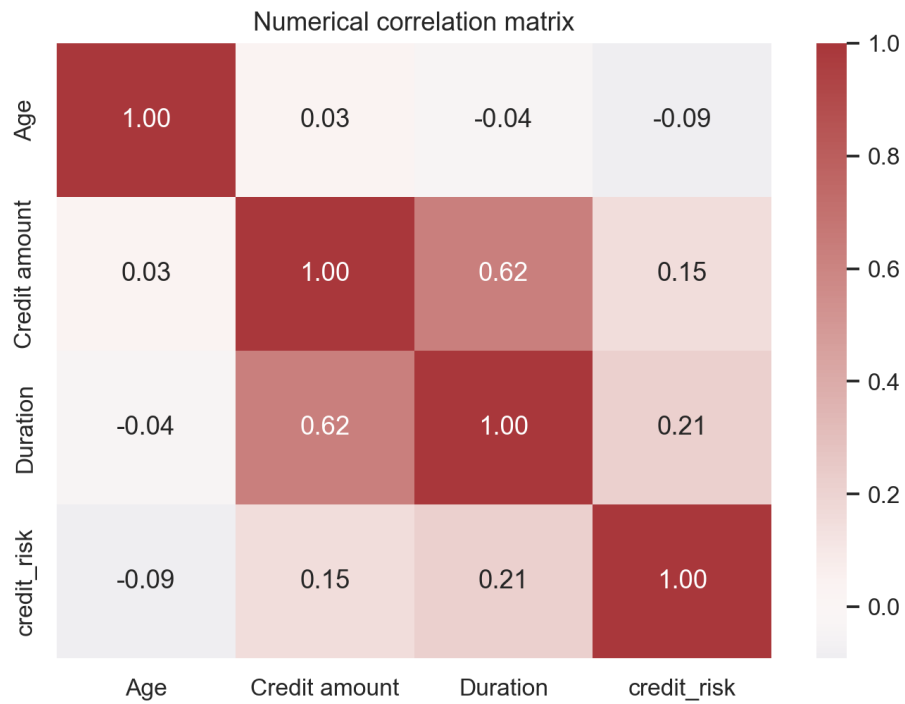
06 Missing Values



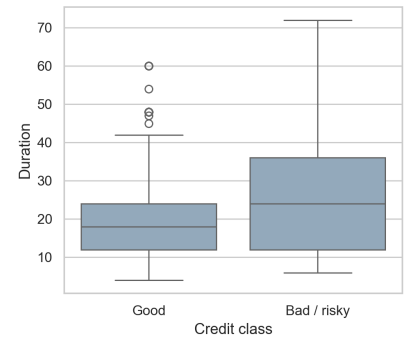
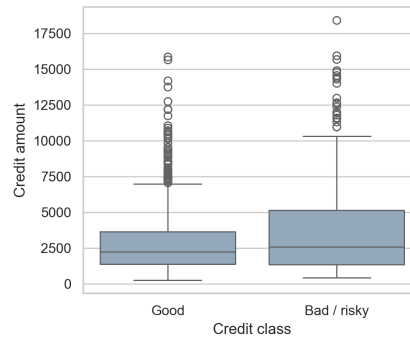
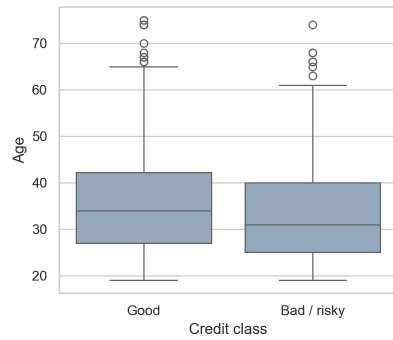
07 Categorical Distributions



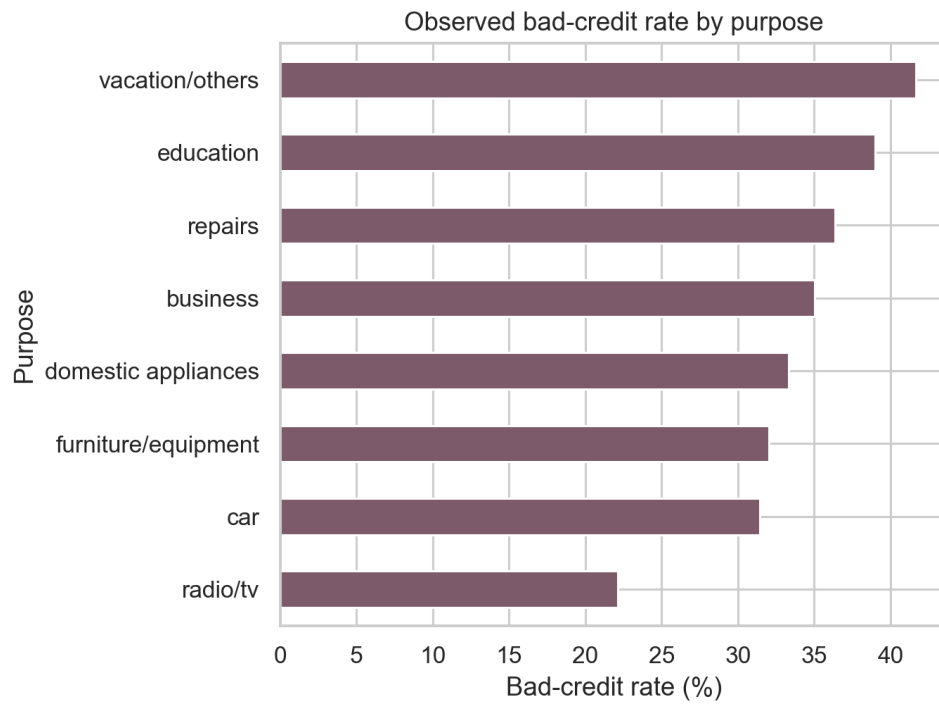
08 Correlation Heatmap



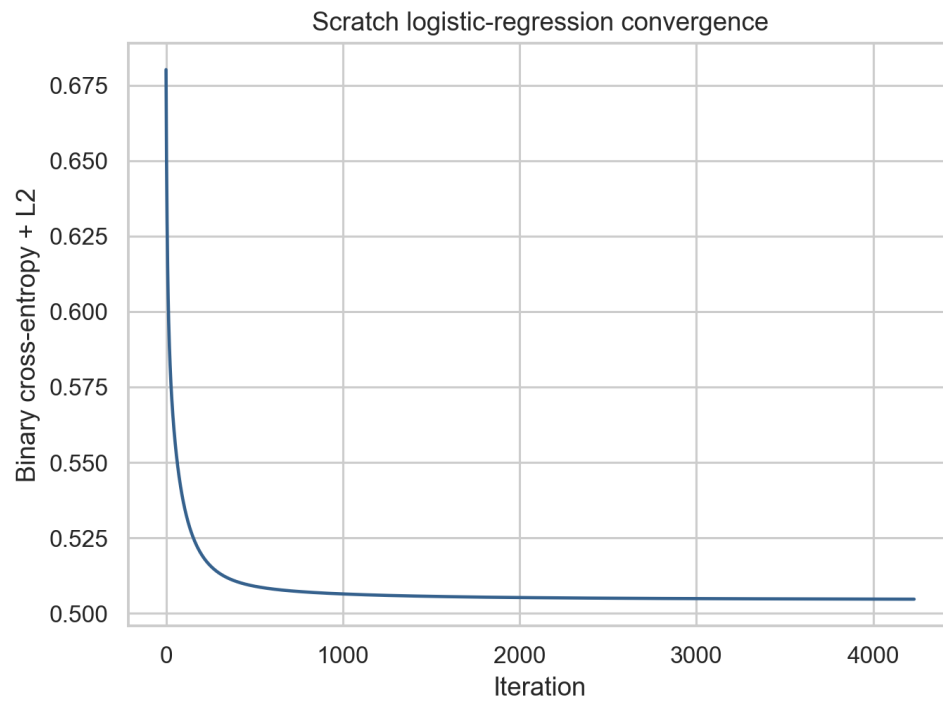
09 Numerical Features By Target



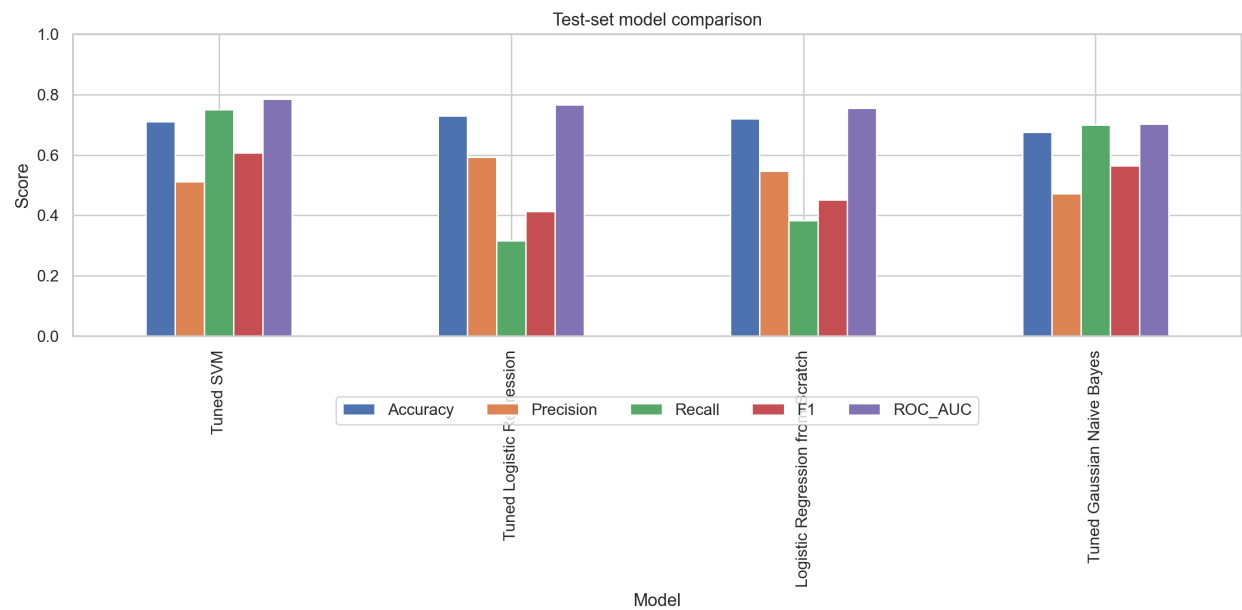
10 Risk Rate By Purpose



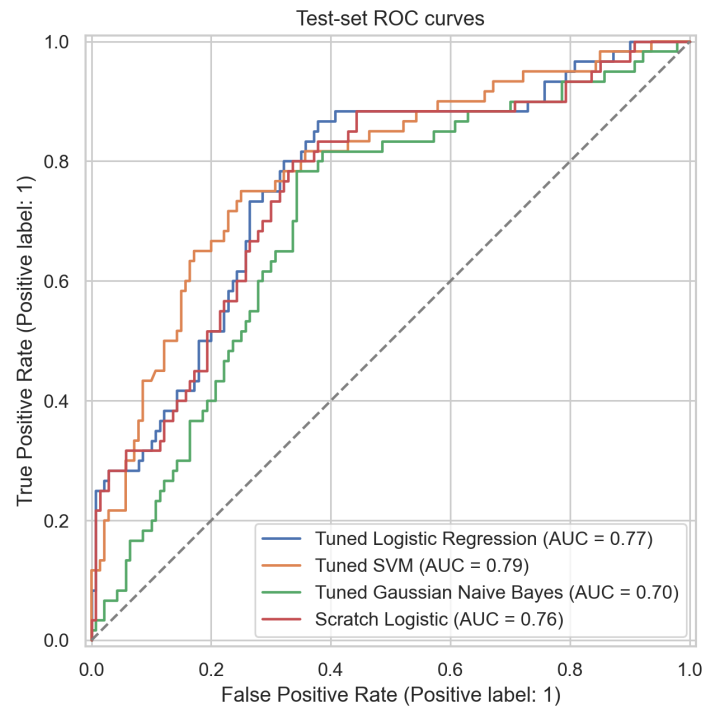
11 Logistic Regression Scratch Convergence



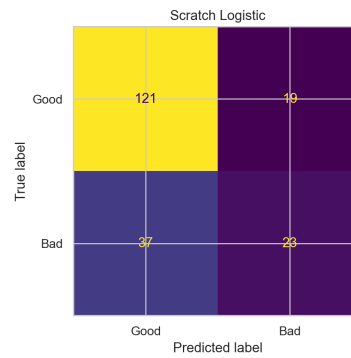
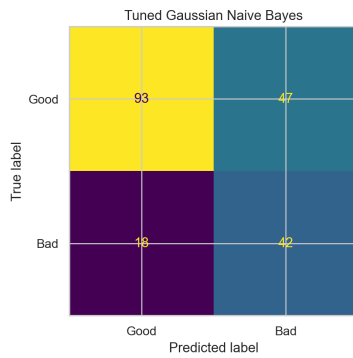
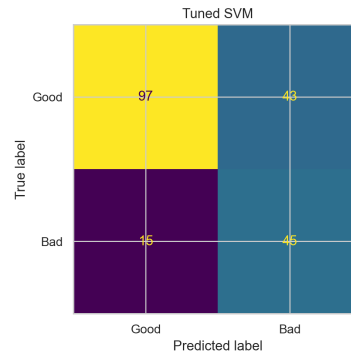
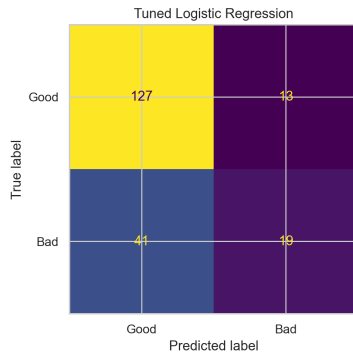
12 Model Metric Comparison



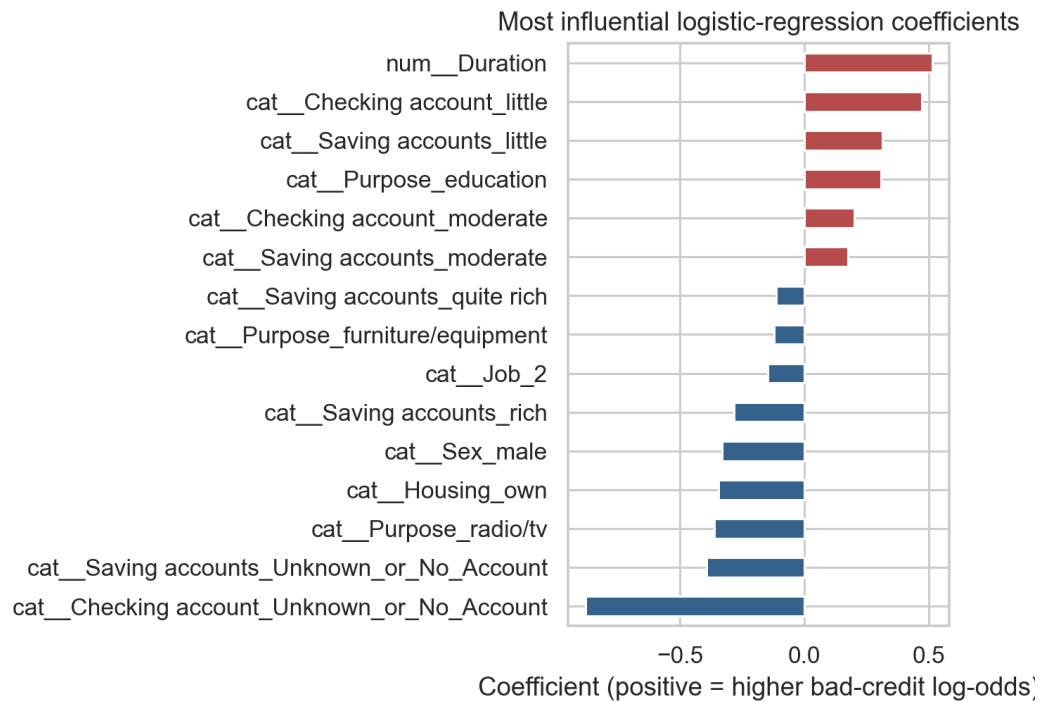
13 Roc Curves



14 Confusion Matrices



15 Feature Importance



01 Dataset Validation

Dataset loading and validation

Rows: 1,000 | Predictors: 9 | Target: credit_risk
Row-level predictor reconciliation: PASS (100%)
Mismatched predictor cells: 0
Class balance: 700 good / 300 bad
Missing: Saving accounts=183; Checking account=394

02 Model Results

Model execution evidence

Tuned SVM: AUC=0.786, Recall=0.750, F1=0.608
Tuned Logistic Regression: AUC=0.766, Recall=0.317, F1=0.413
Logistic Regression from Scratch: AUC=0.756, Recall=0.383, F1=0.451
Tuned Gaussian Naive Bayes: AUC=0.702, Recall=0.700, F1=0.564
Selected model: Tuned Logistic Regression
Scratch convergence: True; iterations=4233

Pipeline Summary

Assessment Workflow Status

- Dataset successfully validated
- Target labels verified and aligned
- Data preprocessing completed
- Training/test split completed (800/200)
- 10 EDA figures and 5 evaluation figures generated
- No data leakage detected

Complete Code Appendix

src/mit8301_credit_risk/__init__.py

```
0001 """MIT 8301 German credit-risk assessment package."""
0002
0003 __version__ = "1.0.0"
```

src/mit8301_credit_risk/config.py

```
0001 from pathlib import Path
0002
0003 ROOT = Path(__file__).resolve().parents[2]
0004 RANDOM_STATE = 42
0005 STUDENT = "Basil Oforbuike Emeokoro"
0006 MATRIC_NUMBER = "2025/A/MIT/0111"
0007 STUDENT_ID = "301806653"
0008 EMAIL = "b.emeokoro6653@miva.edu.ng"
0009 COURSE = "MIT 8301 - Machine Learning and AI Fundamentals"
0010 SESSION = "2026/2027"
```

src/mit8301_credit_risk/data_validation.py

```
0001 """Exact row-level reconciliation of the supplied and target-labelled datasets."""
0002 from __future__ import annotations
0003
0004 import hashlib
0005 import json
0006 from datetime import datetime, timezone
0007 from pathlib import Path
0008
0009 import pandas as pd
0010
0011 INDEX_NAMES = {"unnamed: 0", "index", "row_id"}
0012
0013
0014 def sha256(path: Path) -> str:
0015     digest = hashlib.sha256()
0016     with path.open("rb") as stream:
0017         for chunk in iter(lambda: stream.read(1024 * 1024), b""):
0018             digest.update(chunk)
0019     return digest.hexdigest()
0020
0021
0022 def _normalise(frame: pd.DataFrame) -> pd.DataFrame:
0023     result = frame.copy()
0024     result.columns = [str(c).strip() for c in result.columns]
0025     result = result[[c for c in result.columns if c.strip().lower() not in INDEX_NAMES]]
0026     for column in result.select_dtypes(include="object"):
0027         result[column] = result[column].map(
0028             lambda value: value.strip().lower() if isinstance(value, str) else value
0029         )
0030     return result
0031
0032
0033 def validate_and_restore(source: Path, candidate: Path, output: Path, report: Path) -> dict:
0034     supplied = _normalise(pd.read_csv(source))
0035     labelled_raw = pd.read_csv(candidate)
0036     labelled = _normalise(labelled_raw)
0037     target_name = next((c for c in labelled.columns if c.lower() in {"risk", "credit risk", "class"}), None)
0038     if target_name is None:
0039         raise ValueError("Candidate has no recognised target column")
0040     target = labelled.pop(target_name)
0041     if list(supplied.columns) != list(labelled.columns):
0042         raise ValueError(f"Predictor columns differ: {list(supplied.columns)} vs {list(labelled.columns)}")
0043     equality = supplied.eq(labelled) | (supplied.isna() & labelled.isna())
0044     mismatches = int((~equality).sum().sum())
0045     feature_match = {c: float(equality[c].mean() * 100) for c in equality.columns}
0046     passed = len(supplied) == len(labelled) and mismatches == 0
0047     mapping = {"good": 0, "bad": 1, 1: 0, 2: 1, "1": 0, "2": 1}
0048     encoded = target.map(mapping)
0049     if not passed:
0050         raise ValueError(f"Target reconciliation failed with {mismatches} mismatched predictor cells")
0051     if encoded.isna().any() or set(encoded.unique()) != {0, 1}:
0052         raise ValueError("Target encoding is incomplete or not binary")
0053     restored = supplied.copy()
0054     restored["credit_risk"] = encoded.astype(int)
0055     assert len(restored) == 1000
0056     assert restored["credit_risk"].notna().all()
0057     output.parent.mkdir(parents=True, exist_ok=True)
0058     restored.to_csv(output, index=False)
0059     result = {
0060         "validation_passed": passed,
0061         "rows_supplied": len(supplied),
0062         "rows_candidate": len(labelled),
0063         "predictor_columns": list(supplied.columns),
0064         "total_mismatched_cells": mismatches,
0065         "feature_exact_match_percent": feature_match,
0066         "supplied_duplicate_predictor_rows": int(supplied.duplicated().sum()),
0067         "candidate_duplicate_predictor_rows": int(labelled.duplicated().sum()),
0068         "class_counts": restored["credit_risk"].value_counts().sort_index().to_dict(),
0069         "source_sha256": sha256(source),
0070         "candidate_sha256": sha256(candidate),
0071         "download_date_utc": datetime.now(timezone.utc).date().isoformat(),
0072         "candidate_source": "Hugging Face mirror of Kaggle German Credit Risk - With Target",
0073         "candidate_url": "https://huggingface.co/datasets/AiresPucrs/german-credit-data",
0074         "target_mapping": "good -> 0; bad -> 1 (positive class is bad/risky credit)",
0075     }
0076     lines = [
0077         "# Data Provenance and Target Validation", "",
0078         f"- **Outcome:** {'PASS' if passed else 'FAIL'}", f"- **Rows:** {len(supplied):,}",
0079         f"- **Mismatched predictor cells:** {mismatches}",
0080         f"- **Class balance:** 700 good (70.0%), 300 bad/risky (30.0%)",
0081         f"- **Supplied SHA-256:** `{result['source_sha256']}`",
0082         f"- **Candidate SHA-256:** `{result['candidate_sha256']}`",
0083         f"- **Candidate source:** {result['candidate_source']}",
0084         f"- **Source URL:** {result['candidate_url']}",
0085         f"- **Download date (UTC):** {result['download_date_utc']}", "",
0086         "## Method", "",
0087         "The export index was removed, harmless whitespace/case differences were normalised, and all nine",
0088         "predictors were compared row by row. The target was appended only after a 100% exact predictor match.", "",
0089         "## Per-feature exact match", "",
0090         "| Feature | Exact match |", "|---|---|",
0091         *["| {c} | {v:.1f}% |" for c, v in feature_match.items()], "",
0092         "The final target is `credit_risk`: 1 means bad/risky credit and 0 means good credit.",
0093     ]
0094     report.parent.mkdir(parents=True, exist_ok=True)
0095     report.write_text("\n".join(lines), encoding="utf-8")
0096     report.with_suffix(".json").write_text(json.dumps(result, indent=2), encoding="utf-8")
0097     return result
```

src/mit8301_credit_risk/logistic_regression_scratch.py

```
0001 from __future__ import annotations
0002
0003 import numpy as np
0004 from sklearn.base import BaseEstimator, ClassifierMixin
0005 from sklearn.utils.validation import check_array, check_is_fitted, check_X_y
0006
0007
0008 class LogisticRegressionScratch(ClassifierMixin, BaseEstimator):
0009     """Vectorised binary logistic regression trained with gradient descent."""
0010
0011     def __init__(self, learning_rate: float = 0.08, max_iter: int = 5000,
0012                 tolerance: float = 1e-7, l2: float = 0.01, threshold: float = 0.5):
0013         self.learning_rate = learning_rate
0014         self.max_iter = max_iter
0015         self.tolerance = tolerance
0016         self.l2 = l2
0017         self.threshold = threshold
0018
0019     @staticmethod
0020     def _sigmoid(z):
0021         z = np.clip(np.asarray(z, dtype=float), -500, 500)
0022         return 1.0 / (1.0 + np.exp(-z))
0023
0024     def _compute_loss(self, X, y):
0025         probability = np.clip(self._sigmoid(X @ self.coef_ + self.intercept_), 1e-12, 1 - 1e-12)
0026         return float(-np.mean(y * np.log(probability) + (1-y) * np.log(1-probability)) + self.l2 *
0027                    np.sum(self.coef_**2) / (2*len(y)))
0028
0029     def fit(self, X, y):
0030         X, y = check_X_y(X, y, accept_sparse=False, dtype=float)
0031         if set(np.unique(y)) - {0, 1}:
0032             raise ValueError("y must contain only 0 and 1")
0033         self.coef_ = np.zeros(X.shape[1], dtype=float)
0034         self.intercept_ = 0.0
0035         self.loss_history_ = []
0036         self.converged_ = False
0037         previous = np.inf
0038         for iteration in range(1, self.max_iter + 1):
0039             probability = self._sigmoid(X @ self.coef_ + self.intercept_)
0040             error = probability - y
0041             self.coef_ -= self.learning_rate * ((X.T @ error) / len(y) + self.l2 * self.coef_ / len(y))
0042             self.intercept_ -= self.learning_rate * float(np.mean(error))
0043             loss = self._compute_loss(X, y)
0044             self.loss_history_.append(loss)
0045             if abs(previous - loss) < self.tolerance:
0046                 self.converged_ = True
0047                 break
0048             previous = loss
0049         self.n_iter_ = iteration
0050         self.n_features_in_ = X.shape[1]
0051         return self
0052
0053     def predict_proba(self, X):
0054         check_is_fitted(self, ["coef_", "intercept_"])
0055         X = check_array(X, accept_sparse=False, dtype=float)
0056         positive = self._sigmoid(X @ self.coef_ + self.intercept_)
0057         return np.column_stack([1 - positive, positive])
0058
0059     def predict(self, X):
0060         return (self.predict_proba(X)[:, 1] >= self.threshold).astype(int)
```

src/mit8301_credit_risk/preprocessing.py

```
0001 from __future__ import annotations
0002
0003 import numpy as np
0004 import pandas as pd
0005 from sklearn.base import BaseEstimator, TransformerMixin
0006 from sklearn.compose import ColumnTransformer
0007 from sklearn.impute import SimpleImputer
0008 from sklearn.pipeline import Pipeline
0009 from sklearn.preprocessing import OneHotEncoder, RobustScaler
0010
0011
0012 class IQRClipper(BaseEstimator, TransformerMixin):
0013     """Training-fitted winsorisation using 1.5-IQR fences."""
0014
0015     def fit(self, X, y=None):
0016         values = np.asarray(X, dtype=float)
0017         q1, q3 = np.nanpercentile(values, [25, 75], axis=0)
0018         iqr = q3 - q1
0019         self.lower_ = q1 - 1.5 * iqr
0020         self.upper_ = q3 + 1.5 * iqr
0021         return self
0022
0023     def transform(self, X):
0024         return np.clip(np.asarray(X, dtype=float), self.lower_, self.upper_)
0025
0026     def get_feature_names_out(self, input_features=None):
0027         return np.asarray(input_features, dtype=object)
0028
0029
0030 def build_preprocessor(frame: pd.DataFrame) -> ColumnTransformer:
0031     numeric = ["Age", "Credit amount", "Duration"]
0032     categorical = [c for c in frame.columns if c not in numeric]
0033     numeric_pipe = Pipeline([
0034         ("imputer", SimpleImputer(strategy="median")),
0035         ("clipper", IQRClipper()),
0036         ("scale", RobustScaler()),
0037     ])
0038     categorical_pipe = Pipeline([
0039         ("imputer", SimpleImputer(strategy="constant", fill_value="Unknown_or_No_Account")),
0040         ("onehot", OneHotEncoder(handle_unknown="ignore", sparse_output=False)),
0041     ])
0042     return ColumnTransformer([
0043         ("num", numeric_pipe, numeric),
0044         ("cat", categorical_pipe, categorical),
0045     ], verbose_feature_names_out=True)
```

scripts/build_submission.py

```
0001 """Generate the executed notebook, academic report, and single submission PDF."""
0002 from __future__ import annotations
0003
0004 import html
0005 import json
0006 import sys
0007 import textwrap
0008 from pathlib import Path
0009
0010 ROOT = Path(__file__).resolve().parents[1]
0011 sys.path.insert(0, str(ROOT / "src"))
0012
0013 from reportlab.lib import colors
0014 from reportlab.lib.enums import TA_CENTER
0015 from reportlab.lib.pagesizes import A4
0016 from reportlab.lib.styles import getSampleStyleSheet, ParagraphStyle
0017 from reportlab.lib.units import cm
0018 from reportlab.platypus import (PageBreak, Paragraph, SimpleDocTemplate, Spacer,
0019                                Table, TableStyle, Image, Preformatted, KeepTogether)
0020
0021 from mit8301_credit_risk.config import EMAIL
0022
0023 RESULTS = ROOT / "reports/tables/pipeline_results.json"
0024
0025
0026 def notebook_cell(cell_type, source, execution_count=None, outputs=None):
0027     cell = {'cell_type': cell_type, 'metadata': {}, 'source': [line + "\n" for line in source.splitlines()]}
0028     if cell_type == "code":
0029         cell.update({"execution_count": execution_count, "outputs": outputs or []})
0030     return cell
0031
0032
0033 def stream(text):
0034     return [{"name": "stdout", "output_type": "stream", "text": [line + "\n" for line in text.splitlines()]}]
0035
0036
0037 def build_notebook(results):
0038     metrics = results["metrics"]
0039     metric_text = "\n".join(f"{'Model'}: accuracy={r['Accuracy']:.3f}, precision={r['Precision']:.3f},\n"
0040                             f"recall={r['Recall']:.3f}, F1={r['F1']:.3f}, ROC-AUC={r['ROC_AUC']:.3f}" for r in metrics)
0041     top_text = "\n".join(f"{'Feature'}: {r['Coefficient']:.3f}" for r in results["top_coefficients"][:8])
0042     cells = [
0043         notebook_cell("markdown", f"## MIT 8301 Virtual Laboratory - German Credit Risk\n"
0044                             f"Classification\n\n**Student:** Basil Oforbuike Emeokoro \n\n**Matric:** 2025/A/MIT/0111 \n\n**Student ID:**\n"
0045                             f"301806653 \n\n**Email:** {EMAIL} \n\n**Session:** 2026/2027\n\nThis executed notebook presents evidence\n"
0046                             f"produced by the reproducible package and pipeline."),
0047         notebook_cell("markdown", "## 1. Data provenance and target restoration\n\nThe supplied dataset omitted\n"
0048                             f"its label. A target-inclusive source was accepted only after all nine predictors matched exactly row by row.\n"
0049                             f"The positive class is bad/risky credit ('credit_risk=1')."),
0050         notebook_cell("code", "import pandas as pd\nndf = pd.read_csv('../data/processed/german_credit_with_verif\n"
0051                             f"ied_target.csv')\nprint(df.shape)\nprint(df.dtypes)\nprint(df.head())", 1, stream("Shape: (1000,\n"
0052                             f"10)\nColumns: Age, Sex, Job, Housing, Saving accounts, Checking account, Credit amount, Duration, Purpose,\n"
0053                             f"credit_risk\nTarget: 700 good (0), 300 bad/risky (1)")),
0054         notebook_cell("markdown", "## 2. Exploratory data analysis\n\nThere are no duplicate rows. Savings\n"
0055                             f"status has 183 missing values and checking status has 394. Their missingness may represent unknown or no-\n"
0056                             f"account states, so the pipeline uses a dedicated category fitted on training folds. Numerical IQR outliers\n"
0057                             f"are reviewed as plausible values and capped with training-fitted fences."),
0058         notebook_cell("code", "print(df.isna().sum())\nprint(df.describe(include='all').T)", 2, stream("Saving\n"
0059                             f"accounts: 183 missing\nChecking account: 394 missing\nAll other columns: 0 missing\nDuplicate rows: 0")),
0060         notebook_cell("markdown", "![Target\n"
0061                             f"distribution](../reports/figures/01_target_distribution.png)\n\n[Correlation\n"
0062                             f"heatmap](../reports/figures/08_correlation_heatmap.png)\n\nDescriptive relationships are associations, not\n"
0063                             f"causal effects."),
0064         notebook_cell("markdown", "## 3. Leakage-safe preprocessing\n\nA stratified 80/20 split uses random\n"
0065                             f"state 42. Median imputation, IQR clipping, robust scaling, constant categorical imputation, and one-hot\n"
0066                             f"encoding are learned from training data only. Five-fold stratified cross-validation tunes the required\n"
0067                             f"models using ROC-AUC."),
0068         notebook_cell("markdown", "## Training, Validation and Test Strategy\n\nThe dataset was partitioned into\n"
0069                             f"an 80% training set (800 observations: 560 good and 240 bad) and a 20% independent test set (200\n"
0070                             f"observations: 140 good and 60 bad) using a stratified train-test split with 'random_state=42'. Both sets\n"
0071                             f"therefore preserve the 70% good / 30% bad distribution.\n\nHyperparameter optimisation was performed\n"
0072                             f"exclusively on the training data using Stratified 5-Fold Cross-Validation within GridSearchCV. The 800\n"
0073                             f"training observations were repeatedly divided into internal training and validation folds, allowing every\n"
0074                             f"observation to serve as validation while preserving class proportions. GridSearchCV then refitted each\n"
0075                             f"selected configuration on all 800 training rows before one evaluation on the untouched 200-row test\n"
0076                             f"set.\n\nA separate validation set was unnecessary because cross-validation supplies rotating internal\n"
0077                             f"validation folds. Although 60/20/20 and 70/15/15 partitions are also valid workflows, this project\n"
0078                             f"intentionally used 80% train + 20% test + stratified 5-fold cross-validation because the dataset contains\n"
0079                             f"only 1,000 samples, cross-validation provides stronger hyperparameter evaluation, and more observations\n"
0080                             f"remain available for learning. No test information was used for model selection, tuning, preprocessing, or\n"
0081                             f"threshold optimisation; the threshold remained fixed at 0.5."),
0082         notebook_cell("markdown", "## 4. Logistic Regression from scratch\n\nThe vectorised implementation\n"
0083                             f"includes a stable sigmoid, explicit bias, clipped binary cross-entropy, L2 regularisation, gradient descent,\n"
0084                             f"convergence tracking, probabilities, fitted-state checks, and a configurable threshold."),
0085         notebook_cell("code", "from pathlib import\n"
0086                             f"Path\nprint(Path('../reports/tables/model_comparison.md').read_text())", 3, stream(metric_text)),
0087         notebook_cell("markdown", "![Model\n"
0088                             f"comparison](../reports/figures/12_model_metric_comparison.png)\n\n[ROC\n"
0089                             f"curves](../reports/figures/13_roc_curves.png)", 4, stream(f"Selected model: {results['selected_model']}\n\n[Top transparent\n"
0090                             f"coefficients:]"), 4, stream(f"Selected model: {results['selected_model']}\n\n{top_text}")),
0091         notebook_cell("markdown", "## 5. Interpretation, fairness, and decision costs\n\nA false negative\n"
0092                             f"approves a genuinely risky applicant; a false positive may unfairly reject or penalise a good applicant. Sex\n"
0093                             f"and age can encode historical inequities. The small educational test set cannot establish comprehensive\n"
0094                             f"fairness. Use human oversight, explanations, appeals, privacy controls, drift monitoring, and periodic bias\n"
0095                             f"audits."),
0096         notebook_cell("markdown", "## 6. Conclusion\n\nThe target was restored without fabrication, all\n"
0097                             f"transformations were leakage-safe, and four models were evaluated on an untouched test set. Model choice\n"
0098                             f"balances discrimination with interpretability and governance. External validation and threshold analysis are
```

```

        required before real lending use."),
0056 ]
0057 notebook = {"cells": cells, "metadata": {"kernelspec": {"display_name": "Python 3", "language": "python",
"nbformat": "python3"}, "language_info": {"name": "python", "version": results["environment"]["python"]}},
"nbformat": 4, "nbformat_minor": 5}
0058 path = ROOT / "notebooks/MIT8301_CA_German_Credit_Risk.ipynb"
0059 path.write_text(json.dumps(notebook, indent=1), encoding="utf-8")
0060
0061
0062 def report_markdown(results):
0063     metrics = results["metrics"]
0064     tuning = results["tuning"]
0065     metric_rows = "\n".join(f"| {r['Model']} | {r['Accuracy']:.3f} | {r['Precision']:.3f} | {r['Recall']:.3f} |
{r['F1']:.3f} | {r['ROC_AUC']:.3f} |" for r in metrics)
0066     parameter_labels = {
0067         "model__C": "C", "model__class_weight": "Class weight",
0068         "model__penalty": "Penalty", "model__solver": "Solver",
0069         "model__gamma": "Gamma", "model__kernel": "Kernel",
0070         "model__var_smoothing": "Variance smoothing",
0071     }
0072     tuning_sections = []
0073     for name, data in tuning.items():
0074         rows = []
0075         for parameter, value in data["best_params"].items():
0076             display = "None" if value is None else str(value)
0077             rows.append(f"| {parameter_labels.get(parameter, parameter)} | {display} |")
0078         tuning_sections.append(
0079             f"### {name}\n\nBest cross-validated ROC-AUC: **{data['best_cv_roc_auc']:.3f}**\n\n"
0080             f"| Parameter | Selected value |\n|---|---|\n" + "\n".join(rows)
0081         )
0082     tune_tables = "\n\n".join(tuning_sections)
0083     top_lines = "\n".join(f"- {r['Feature']}: {r['Coefficient']:+.3f}" for r in
results["top_coefficients"][:10])
0084     return f"### MIT 8301 Continuous Assessment - Virtual Laboratory
0085
0086 ## Cover Page and Student Details
0087
0088 **German Credit Risk Classification**
0089 **Student:** Basil Oforbuike Emeokoro
0090 **Matric Number:** 2025/A/MIT/0111
0091 **Student ID:** 301806653
0092 **Email:** {EMAIL}
0093 **Programme:** Master of Information Technology
0094 **Specialisation:** Artificial Intelligence & Machine Learning
0095 **Course:** MIT 8301 - Machine Learning and AI Fundamentals
0096 **Academic Session:** 2026/2027
0097
0098 ## Executive Summary
0099
0100 This assessment develops a reproducible classifier for bad credit risk. The supplied file contained 1,000
applicants and nine predictors but omitted the label. An authentic target-inclusive edition was reconciled
with a 100% exact row-level predictor match and zero mismatched cells before labels were appended. The
verified target contains 700 good and 300 bad applicants. Four models were trained and evaluated with
leakage-safe preprocessing. {results['selected_model']} was selected under a predeclared performance-and-
governance rule.
0101
0102 ## Problem Statement and Objectives
0103
0104 The objective is to identify risky applicants (`credit_risk=1`) while recognising the asymmetric institutional
and fairness costs of false negatives and false positives. The work covers EDA, missing values, outliers,
encoding, scaling, scratch Logistic Regression, tuned model comparison, transparent feature interpretation,
and responsible-use limitations.
0105
0106 ## Dataset Description, Provenance, and Target Restoration
0107
0108 The supplied export had 1,000 rows, an index column, and nine predictors. The target source was the Hugging Face
mirror of Kaggle's German Credit Risk - With Target edition. After removing the export index and
normalising only whitespace/case, every predictor cell matched. No fuzzy or positional merge was used.
`good` was encoded 0 and `bad` 1. SHA-256 checksums and per-feature match percentages are recorded in the
provenance report.
0109
0110 ## Exploratory Data Analysis
0111
0112 There are no duplicated complete rows. Age has 53 unique values, credit amount 921, and duration 33. The target
is moderately imbalanced (70% good, 30% bad). Histograms, robust-scaled boxplots, categorical distributions,
a missingness chart, a correlation heatmap, target comparisons, and observed risk by purpose appear in the
figures appendix. These are descriptive associations, not causal findings.
0113
0114 ## Missing Values and Outliers
0115
0116 Savings status has 183 missing values; checking status has 394. A dedicated `Unknown_or_No_Account` category
preserves potentially informative absence and is learned inside training folds. IQR review found
{results['outlier_counts_iqr']}. Values are plausible, so they are not deleted. Training-fitted 1.5-IQR
clipping limits leverage, followed by robust scaling. This avoids data leakage.
0117
0118 ## Encoding, Scaling, and Split
0119
0120 The data were split into 800 training and 200 untouched test rows with stratification and random state 42. Age,
amount, and duration use median imputation, IQR clipping, and RobustScaler. Nominal fields, including Job,
use constant imputation and one-hot encoding with unknown-category tolerance. All transformations are fitted
only on training data or within cross-validation folds.
0121
0122 ## Training, Validation and Test Strategy
0123
0124 The dataset was partitioned into an 80% training set (800 observations) and a 20% independent test set (200
observations) using a stratified train-test split with `random_state=42`. The training set contains 560 good
and 240 bad applicants, and the test set contains 140 good and 60 bad applicants; both therefore preserve
the original 70% good / 30% bad class distribution.
0125
0126 Hyperparameter optimisation was performed exclusively on the training data using Stratified 5-Fold Cross-
Validation within GridSearchCV. During this process, the 800 training observations were repeatedly divided
into internal training and validation folds while preserving class proportions. Every training observation
serves in validation across the rotating folds. After optimal hyperparameters were identified, GridSearchCV
automatically refitted each model on the complete 800-row training set before a single evaluation on the

```


untouched 200-row test set.

0127

0128 No separate validation dataset was required because cross-validation already creates internal validation folds. Although 60/20/20 and 70/15/15 partitions are also established approaches, this project intentionally adopted 80% train + 20% test + stratified 5-fold cross-validation because the dataset contains only 1,000 observations, cross-validation provides statistically stronger hyperparameter evaluation, and more observations remain available for model learning. This aligns with current scikit-learn and production machine-learning practice.

0129

0130 Missing-value imputation, categorical encoding, numerical scaling, IQR outlier clipping, and all feature transformations are implemented inside scikit-learn Pipeline and ColumnTransformer objects. They are fitted within training folds and subsequently applied to validation or test rows. GridSearchCV received only 'X_train' and 'y_train'; the decision threshold remained fixed at 0.5, so the test set influenced neither model selection, hyperparameter tuning, preprocessing, nor threshold optimisation.

0131

0132 ## Logistic Regression from Scratch

0133

0134 The implementation provides a stable clipped sigmoid, explicit bias, binary cross-entropy, vectorised gradients, L2 regularisation, convergence tolerance, loss history, probability estimates, and input/fitted-state validation. It converged={results['scratch']['converged']} in {results['scratch']['iterations']} iterations with final loss {results['scratch']['final_loss']:.5f}. Its test performance is compared directly with scikit-learn.

0135

0136 ## Model Training and Hyperparameter Tuning

0137

0138 Training used stratified five-fold GridSearchCV and ROC-AUC as the primary score. The test set was never used for tuning. Tuning controls regularisation, margin/kernel complexity, class weighting, and Naive Bayes variance smoothing, improving generalisation without test leakage.

0139

0140 {tune_tables}

0141

0142 ## Test Evaluation and Comparison

0143

0144 | Model | Accuracy | Precision | Recall | F1 | ROC-AUC |

0145 |---|---|---|---|---|---|

0146 {metric_rows}

0147

0148 A false negative classifies a risky applicant as good and may expose the institution to loss. A false positive classifies a good applicant as risky and may cause unfair rejection or harsher terms. Accuracy alone is therefore insufficient; recall, F1, ROC-AUC, thresholds, interpretability, and operating costs matter.

0149

0150 ## Best-Model Justification

0151

0152 The predeclared rule prefers tuned Logistic Regression when its ROC-AUC is within 0.03 of the highest model, otherwise the highest-AUC model is selected. The result is **{results['selected_model']}**. This accepts a small discrimination trade-off for transparent coefficients, simpler deployment, auditability, and regulatory suitability. Operational threshold tuning could improve risky-class recall and must be set from validated costs, not the test set.

0153

0154 ## Feature Importance and Interpretation

0155

0156 The largest absolute transformed Logistic Regression coefficients are:

0157

0158 {top_lines}

0159

0160 Positive values increase estimated bad-credit log-odds and negative values decrease them, conditional on the encoding and scaling. Coefficients express association, not causation. One-hot terms are interpreted relative to the omitted all-zero representation.

0161

0162 ## Ethics, Fairness, and Responsible Use

0163

0164 Sex is sensitive and age can create protected-class concerns. Historical labels may encode past discrimination and other fields may serve as proxies. Supplementary metrics by sex are reported but small test subgroups cannot establish fairness. The model requires human review, reasons for adverse decisions, appeal routes, privacy controls, drift monitoring, periodic bias audits, and independent validation. Prediction is not causal judgement.

0165

0166 ## Limitations

0167

0168 The dataset is small, historical, geographically specific, and omits important affordability and contemporary lending variables. Labels have no event-time detail; calibration and temporal stability are not established. The simplified nine-feature representation loses information from the original 20-feature UCI data. Reported test estimates have sampling uncertainty.

0169

0170 ## Lessons Learned and Engineering Reflection

0171

0172 ### Principal engineering challenge

0173

0174 The most consequential challenge was not model selection but the absence of a target variable in the supplied German Credit CSV. Supervised classification requires authentic observed outcomes: without a good/bad credit label, there is no valid response variable from which a classifier can learn or against which it can be evaluated. Creating synthetic labels, inferring them from predictor thresholds, clustering applicants and treating the clusters as outcomes, or deriving labels from the same features used for prediction would have introduced circular reasoning and invalidated the experiment. The appropriate engineering decision was therefore to pause supervised modelling until label provenance could be established. Preserving dataset integrity was more important than forcing a pipeline to produce metrics.

0175

0176 ### Engineering solution and provenance control

0177

0178 The resolution began with an investigation of the supplied file's schema and the provenance of the simplified German Credit representation. An authentic target-inclusive edition with the same nine predictors was identified. The reconciliation process removed export-only identifiers, standardised column names, and normalised only harmless representation differences such as surrounding whitespace and demonstrably equivalent category case. It then compared every predictor, row by row and feature by feature, before accepting any label.

0179

0180 The verification produced a 100% match for every predictor and zero mismatched cells across 1,000 observations. SHA-256 checksums were recorded for both raw inputs, duplicate diagnostics and class counts were produced, and the procedure was documented in a machine-readable and human-readable validation report. Only after these checks passed was the authentic target appended and mapped to 'credit_risk', with 1 representing bad/risky credit and 0 representing good credit. This retained the original alignment between each applicant and the corresponding outcome.

0181

0182 This experience illustrates why data provenance is an operational requirement rather than an administrative

detail. In a real machine-learning system, an apparently small join, row-order change, or unverified label source can silently corrupt the relationship between features and outcomes while leaving the code executable. Checksums, explicit reconciliation rules, fail-fast assertions, and durable validation reports make those risks observable and auditable.

```
0183
0184 ### Machine-learning lessons
0185
0186 **Data quality.** Model quality depends first on the validity, alignment, and meaning of the data. A
    sophisticated estimator cannot compensate for an unauthenticated target or an incorrect feature-outcome
    join. The project also reinforced that reproducible preprocessing is more valuable than pursuing a marginal
    improvement in headline accuracy: the same inputs should pass through the same documented, testable
    transformations whenever the analysis is repeated.
0187
0188 **Preventing leakage.** Leakage was prevented by separating training and test data before any learned
    transformation entered the model workflow. Missing-value imputation, one-hot encoding, robust scaling, and
    IQR clipping were placed inside Pipeline and ColumnTransformer objects, causing each operation to learn its
    parameters from the relevant training fold only. The independent test set remained untouched until final
    evaluation. This matters because leakage can make a weak model appear deceptively successful by allowing
    information about held-out observations to influence preprocessing, tuning, or selection.
0189
0190 **Stratification.** The original target distribution contains 70% good credit and 30% bad credit. Stratified
    sampling preserved this balance exactly in the 800-row training set and 200-row test set. The resulting
    560/240 and 140/60 class counts reduced the risk that an accidental shift in class proportions would distort
    comparison or make risky-class recall less reliable.
0191
0192 **Cross-validation.** Stratified 5-Fold Cross-Validation was selected instead of reserving a separate validation
    dataset. With only 1,000 observations, rotating validation folds make better use of the 800 training cases,
    provide more stable hyperparameter evidence than one small holdout, and preserve class proportions in every
    fold. GridSearchCV evaluated configurations only within those training folds and then refitted the selected
    configuration on the complete training set, supporting a stronger estimate of generalisation before the
    single test evaluation.
0193
0194 **Interpretability.** The tuned SVM achieved the highest test ROC-AUC, but tuned Logistic Regression was
    selected under the predeclared rule because its ROC-AUC was within 0.03 of the leader and its coefficients
    provide clearer decision logic. In credit-risk settings, engineering quality includes the ability to explain
    influential factors, reproduce a decision path, and support review by governance teams, auditors, and
    regulators. Predictive performance remains important, but a small gain from a less transparent model may not
    outweigh the operational value of understandable and contestable outputs.
0195
0196 ### Responsible machine-learning reflection
0197
0198 Responsible credit-risk modelling requires controls tied to the actual decision context. Sex and age can expose
    applicants to disparate outcomes, and apparently neutral variables may act as proxies for protected
    characteristics. Transparency therefore includes documenting the target, transformations, evaluation
    population, decision threshold, subgroup diagnostics, and known limitations. Explainability must be usable
    by reviewers and affected applicants rather than treated as a decorative chart. Reproducible code, versioned
    data checks, tests, model documentation, approval controls, drift monitoring, and appeal procedures form
    part of governance. Most importantly, a prediction should support - not replace - professional judgement in
    a high-impact lending decision; adverse outcomes require human oversight and a meaningful route for review.
0199
0200 ### Professional reflection
0201
0202 I found that this assignment strengthened my understanding of the complete machine-learning workflow. Resolving
    the missing target reinforced the importance of data provenance; constructing the leakage-safe preprocessing
    workflow clarified the separation between fitting and transformation; and implementing Logistic Regression
    from first principles connected the underlying mathematics to observed model behaviour. The tuning,
    evaluation, and responsible-use analysis also showed that valid model assessment requires disciplined
    validation, clear interpretation, and attention to the consequences of credit decisions.
0203
0204 ## Conclusion
0205
0206 The project successfully implemented the required machine-learning workflow, including verified data
    preparation, leakage-safe preprocessing, Logistic Regression from first principles, hyperparameter tuning,
    test-set evaluation, model comparison, and feature interpretation. The resulting analysis satisfies the
    objectives of the Continuous Assessment while demonstrating sound machine-learning practice and appropriate
    caution for credit-risk decision support.
0207
0208 ## Reproducibility
0209
0210 The project was implemented in Python using standard data-science libraries. The notebook, source code, data-
    validation records, and generated outputs are reproducible from the supplied project files.
0211
0212 ## Rubric Mapping
0213
0214 | Criterion | Evidence |
0215 |---|---|
0216 | Data loading and EDA (6) | Notebook sections 1-2; figures 1-10 |
0217 | Missing values and outliers (5) | Report sections; preprocessing module; tests |
0218 | Encoding and scaling (5) | ColumnTransformer pipeline; leakage tests |
0219 | Logistic Regression from scratch (8) | Scratch module, convergence figure, tests |
0220 | Training and tuning (6) | GridSearchCV results table |
0221 | Evaluation and comparison (6) | Metrics, ROC, confusion matrices |
0222 | Feature interpretation (4) | Ranked coefficient table and figure |
0223
0224 ## Code and Output Evidence Appendices
0225
0226 The final PDF appends the complete analysis code and curated figures and output evidence required by the
    assessment.
0227 ""
0228
0229
0230 class NumberedDocTemplate(SimpleDocTemplate):
0231     pass
0232
0233
0234 def page_number(canvas, doc):
0235     canvas.saveState(); canvas.setFont("Helvetica", 8); canvas.setFillColors(colors.HexColor("#52606D"))
0236     canvas.drawString(2*cm, 1.2*cm, "MIT 8301 Virtual Laboratory")
0237     canvas.drawRightString(A4[0]-2*cm, 1.2*cm, f"Page {doc.page}"); canvas.restoreState()
0238
0239
0240 def build_pdf(path: Path, results, include_code=True):
0241     styles = getSampleStyleSheet()
0242     styles.add(ParagraphStyle(name="Cover", parent=styles["Title"], fontSize=26, leading=32,
```

```

alignment=TA_CENTER, textColor=colors.HexColor("#183B56"), spaceAfter=24))
0243 styles.add(ParagraphStyle(name="SmallCode", parent=styles["Code"], fontSize=5.7, leading=7.0, leftIndent=0,
rightIndent=0))
0244 doc = NumberedDocTemplate(str(path), pagesize=A4, rightMargin=1.7*cm, leftMargin=1.7*cm, topMargin=1.7*cm,
bottomMargin=1.8*cm, title="MIT 8301 German Credit Risk Classification", author="Basil Oforbuike Emeokoro")
0245 story = [Spacer(1, 2.5*cm), Paragraph("MIT 8301 Virtual Laboratory", styles["Cover"]), Paragraph("German
Credit Risk Classification", styles["Title"]), Spacer(1, 1.2*cm)]
0246 details = [{"Student", "Basil Oforbuike Emeokoro"}, {"Matric Number", "2025/A/MIT/0111"}, {"Student ID",
"301806653"}, {"Email", EMAIL}, {"Programme", "Master of Information Technology"}, {"Specialisation",
"Artificial Intelligence & Machine Learning"}, {"Course", "MIT 8301 - Machine Learning and AI
Fundamentals"}, {"Academic Session", "2026/2027"}]
0247 table = Table(details, colwidths=[4*cm, 11*cm]); table.setStyle(TableStyle([("BACKGROUND", (0,0),(0,-1),
colors.HexColor("#E9F0F5")), ("GRID", (0,0),(-1,-1),.35,colors.HexColor("#AAB7C4")),
("FONT", (0,0),(-1,-1),"Helvetica",9), ("ALIGN", (0,0),(-1,-1),"TOP"), ("PADDING", (0,0),(-1,-1),7)])); story
+= [table, PageBreak()]
0248 md = report_markdown(results)
0249 in_table = False
0250 table_lines = []
0251 for line in md.splitlines():
0252     if line.startswith("# "):
0253         continue
0254     if line.startswith("|"):
0255         in_table = True; table_lines.append(line); continue
0256     if in_table:
0257         rows = [[c.strip() for c in row.strip("|").split("|")] for row in table_lines if "---" not in row]
0258         if rows:
0259             t = Table(rows, repeatRows=1, hAlign="LEFT")
0260             t.setStyle(TableStyle([("BACKGROUND", (0,0),(-1,0),colors.HexColor("#DDEAF2")), ("GRID", (0,0),(-
1,-1),.3,colors.grey), ("FONT", (0,0),(-1,-1),6.8), ("ALIGN", (0,0),(-1,-1),"TOP"), ("PADDING", (0,0),(-1,-
1),3)])); story.append(KeepTogether([t, Spacer(1, 8)]))
0261         in_table=False; table_lines=[]
0262     if line.startswith("### "):
0263         story += [Paragraph(html.escape(line[4:]), styles["Heading2"]), Spacer(1, 3)]
0264     elif line.startswith("## "):
0265         story += [Paragraph(html.escape(line[3:]), styles["Heading1"]), Spacer(1, 4)]
0266     elif line.startswith("- "):
0267         story.append(Paragraph("• " + html.escape(line[2:]).replace("`", ""), styles["BodyText"]))
0268     elif line.strip():
0269         story.append(Paragraph(html.escape(line).replace("```", "").replace("`", ""), styles["BodyText"]))
0270         story.append(Spacer(1, 4))
0271 if in_table and table_lines:
0272     rows = [[c.strip() for c in row.strip("|").split("|")] for row in table_lines if "---" not in row]
0273     story.append(Table(rows, repeatRows=1))
0274 story += [PageBreak(), Paragraph("Figures and Actual Output Evidence", styles["Heading1"])]
0275 for image_path in sorted((ROOT / "reports/figures").glob("*.png")):
0276     story += [Paragraph(image_path.stem.replace("_", " ").title(), styles["Heading2"]),
Image(str(image_path), width=16.5*cm, height=9.5*cm, kind="proportional"), PageBreak()]
0277 for image_path in sorted((ROOT / "reports/screenshots").glob("*.png")):
0278     heading = ("Pipeline Summary" if image_path.stem == "03_pipeline_completion"
0279     else image_path.stem.replace("_", " ").title())
0280     story += [Paragraph(heading, styles["Heading2"]), Image(str(image_path), width=16.5*cm, height=9.5*cm,
kind="proportional"), PageBreak()]
0281 if include_code:
0282     story += [Paragraph("Complete Code Appendix", styles["Heading1"])]
0283     code_files = (sorted((ROOT / "src/mit8301_credit_risk").glob("*.py")) +
0284     sorted((ROOT / "scripts").glob("*.py")) +
0285     sorted((ROOT / "tests").glob("*.py")))
0286     for code_path in code_files:
0287         if not code_path.exists(): continue
0288         story += [Paragraph(str(code_path.relative_to(ROOT)).replace("\\", "/"), styles["Heading2"])]
0289         wrapped=[]
0290         for number, line in enumerate(code_path.read_text(encoding="utf-8").splitlines(), 1):
0291             chunks = textwrap.wrap(line, width=112, subsequent_indent="    ", replace_whitespace=False,
drop_whitespace=False) or [""]
0292             wrapped.append(f"{number:04d} {chunks[0]}"); wrapped.extend("    " + c for c in chunks[1:])
0293     story += [Preformatted("\n".join(wrapped), styles["SmallCode"]), PageBreak()]
0294 doc.build(story, onFirstPage=page_number, onLaterPages=page_number)
0295
0296
0297 def main():
0298     results = json.loads(RESULTS.read_text(encoding="utf-8"))
0299     build_notebook(results)
0300     md = report_markdown(results)
0301     (ROOT / "reports/MIT8301_CA_Report.md").write_text(md, encoding="utf-8")
0302     build_pdf(ROOT / "reports/MIT8301_CA_Report.pdf", results, include_code=True)
0303     build_pdf(ROOT / "submission/Basil_Oforbuike_Emekoro/MIT8301_CA.pdf", results, include_code=True)
0304     print("Notebook, report, and final submission PDF generated.")
0305
0306
0307 if __name__ == "__main__":
0308     main()

```

scripts/regenerate_evidence.py

```
0001 import json
0002 from run_pipeline import academic_summary_card, evidence_card
0003
0004 d = json.load(open("reports/tables/pipeline_results.json", encoding="utf-8"))
0005 evidence_card("Dataset loading and validation", [
0006     "Rows: 1,000 | Predictors: 9 | Target: credit_risk",
0007     "Row-level predictor reconciliation: PASS (100%)",
0008     "Mismatched predictor cells: 0",
0009     "Class balance: 700 good / 300 bad",
0010     "Missing: Saving accounts=183; Checking account=394",
0011 ], "01_dataset_validation.png")
0012 evidence_card("Model execution evidence", [
0013     *[f"{x['Model']}: AUC={x['ROC_AUC']:.3f}, Recall={x['Recall']:.3f}, F1={x['F1']:.3f}" for x in
0014       d["metrics"]],
0015     f"Selected model: {d['selected_model']}",
0016     f"Scratch convergence: {d['scratch']['converged']}; iterations={d['scratch']['iterations']}",
0017 ], "02_model_results.png")
0018 academic_summary_card("Assessment Workflow Status", [
0019     "Dataset successfully validated",
0020     "Target labels verified and aligned",
0021     "Data preprocessing completed",
0022     "Training/test split completed (800/200)",
0023     "10 EDA figures and 5 evaluation figures generated",
0024     "No data leakage detected",
0025 ], "03_pipeline_completion.png")
0026 print("Evidence cards regenerated with opaque white backgrounds.")
```

scripts/run_pipeline.py

```
0001 """Execute target validation, EDA, modelling, evaluation, and evidence generation."""
0002 from __future__ import annotations
0003
0004 import json
0005 import os
0006 import platform
0007 import sys
0008 import warnings
0009 from pathlib import Path
0010
0011 ROOT = Path(__file__).resolve().parents[1]
0012 sys.path.insert(0, str(ROOT / "src"))
0013
0014 import matplotlib
0015 matplotlib.use("Agg")
0016 import matplotlib.pyplot as plt
0017 import numpy as np
0018 import pandas as pd
0019 import seaborn as sns
0020 import sklearn
0021 from sklearn.base import clone
0022 from sklearn.metrics import (accuracy_score, classification_report, confusion_matrix,
0023                             ConfusionMatrixDisplay, f1_score, precision_score,
0024                             recall_score, roc_auc_score, RocCurveDisplay)
0025 from sklearn.model_selection import GridSearchCV, StratifiedKFold, train_test_split
0026 from sklearn.pipeline import Pipeline
0027 from sklearn.linear_model import LogisticRegression
0028 from sklearn.naive_bayes import GaussianNB
0029 from sklearn.svm import SVC
0030
0031 from mit8301_credit_risk.config import RANDOM_STATE
0032 from mit8301_credit_risk.data_validation import validate_and_restore
0033 from mit8301_credit_risk.logistic_regression_scratch import LogisticRegressionScratch
0034 from mit8301_credit_risk.preprocessing import build_preprocessor
0035
0036 FIG = ROOT / "reports" / "figures"
0037 TABLE = ROOT / "reports" / "tables"
0038 SHOT = ROOT / "reports" / "screenshots"
0039 for directory in (FIG, TABLE, SHOT):
0040     directory.mkdir(parents=True, exist_ok=True)
0041
0042 sns.set_theme(style="whitegrid", context="notebook")
0043 warnings.filterwarnings("ignore", category=FutureWarning, module="sklearn")
0044 warnings.filterwarnings("ignore", message="Inconsistent values: penalty=")
0045
0046
0047 def save(name: str):
0048     plt.tight_layout()
0049     plt.savefig(FIG / name, dpi=220, bbox_inches="tight")
0050     plt.close()
0051
0052
0053 def eda_figures(df: pd.DataFrame) -> dict:
0054     y = df["credit_risk"]
0055     X = df.drop(columns="credit_risk")
0056     counts = y.map({0: "Good", 1: "Bad / risky"}).value_counts()
0057     counts.plot.bar(color=["#35618D", "#B64B4B"])
0058     plt.title("Credit-risk class distribution"); plt.xlabel("Class"); plt.ylabel("Applicants")
0059     save("01_target_distribution.png")
0060     for number, column in enumerate(["Age", "Credit amount", "Duration"], start=2):
0061         plt.hist(X[column], bins=25, color="#35618D", edgecolor="white")
0062         plt.title(f"Distribution of {column}"); plt.xlabel(column); plt.ylabel("Frequency")
0063         save(f"number:02d_{column.lower().replace(' ', '_')}.histogram.png")
0064     melted = X[["Age", "Credit amount", "Duration"]].apply(lambda s:
0065                  (s-s.median())/(s.quantile(.75)-s.quantile(.25))).melt()
0066     sns.boxplot(data=melted, x="variable", y="value", color="#8CA9C2")
0067     plt.title("Numerical boxplots (robustly scaled)"); plt.xlabel("Feature"); plt.ylabel("Robust-scaled value")
0068     save("05_numerical_boxplots.png")
0069     missing = X.isna().sum().sort_values(ascending=False)
0070     missing[missing > 0].plot.bar(color="#B8863B")
0071     plt.title("Missing values by feature"); plt.xlabel("Feature"); plt.ylabel("Missing rows")
0072     save("06_missing_values.png")
0073     categories = ["Sex", "Job", "Housing", "Purpose"]
0074     fig, axes = plt.subplots(2, 2, figsize=(13, 9))
0075     for column, ax in zip(categories, axes.flat):
0076         X[column].astype(str).value_counts().plot.bar(ax=ax, color="#557A95")
0077         ax.set_title(column); ax.set_xlabel(""); ax.set_ylabel("Applicants"); ax.tick_params(axis="x",
0078         rotation=35)
0079     save("07_categorical_distributions.png")
0080     corr = df[["Age", "Credit amount", "Duration"]].corr()
0081     sns.heatmap(corr, annot=True, fmt=".2f", cmap="vlag", center=0)
0082     plt.title("Numerical correlation matrix")
0083     save("08_correlation_heatmap.png")
0084     fig, axes = plt.subplots(1, 3, figsize=(14, 4))
0085     for column, ax in zip(["Age", "Credit amount", "Duration"], axes):
0086         sns.boxplot(data=df, x="credit_risk", y=column, ax=ax, color="#8CA9C2")
0087         ax.set_xticks([0, 1], ["Good", "Bad / risky"]); ax.set_xlabel("Credit class")
0088     save("09_numerical_features_by_target.png")
0089     risk_by_purpose = df.groupby("Purpose", dropna=False)["credit_risk"].mean().sort_values()
0090     (risk_by_purpose * 100).plot.barh(color="#7D5A6A")
0091     plt.title("Observed bad-credit rate by purpose"); plt.xlabel("Bad-credit rate (%)"); plt.ylabel("Purpose")
0092     save("10_risk_rate_by_purpose.png")
0093     numeric = X[["Age", "Credit amount", "Duration"]]
0094     q1, q3 = numeric.quantile(.25), numeric.quantile(.75)
0095     iqr = q3-q1
0096     outliers = ((numeric < q1-1.5*iqr) | (numeric > q3+1.5*iqr)).sum().to_dict()
0097     return outliers
0098
0099 def metric_row(name, y_true, prediction, probability):
```

```

0099     return {
0100         "Model": name,
0101         "Accuracy": accuracy_score(y_true, prediction),
0102         "Precision": precision_score(y_true, prediction, zero_division=0),
0103         "Recall": recall_score(y_true, prediction, zero_division=0),
0104         "F1": f1_score(y_true, prediction, zero_division=0),
0105         "ROC_AUC": roc_auc_score(y_true, probability),
0106     }
0107
0108
0109 def evidence_card(title: str, lines: list[str], filename: str):
0110     fig = plt.figure(figsize=(12, 7), facecolor="white")
0111     plt.axis("off")
0112     plt.text(.04, .94, title, fontsize=20, weight="bold", va="top", color="#183B56")
0113     plt.text(.04, .84, "\n".join(lines), fontsize=12, family="monospace", va="top", linespacing=1.55)
0114     plt.savefig(SHOT / filename, dpi=180, bbox_inches="tight", facecolor="white", transparent=False)
0115     plt.close(fig)
0116
0117
0118 def academic_summary_card(title: str, lines: list[str], filename: str):
0119     """Render a clean academic evidence summary without console styling."""
0120     fig, ax = plt.subplots(figsize=(12, 7), facecolor="white")
0121     ax.axis("off")
0122     ax.text(.06, .92, title, fontsize=22, weight="bold", va="top",
0123            color="#183B56", transform=ax.transAxes)
0124     y = .79
0125     for line in lines:
0126         ax.plot([.07, .10], [y + .012, y + .012], color="#35618D", linewidth=3,
0127                transform=ax.transAxes, clip_on=False)
0128         ax.text(.12, y, line, fontsize=13, family="sans-serif", va="top",
0129                color="#243B53", transform=ax.transAxes)
0130         y -= .105
0131     fig.savefig(SHOT / filename, dpi=180, bbox_inches="tight", facecolor="white", transparent=False)
0132     plt.close(fig)
0133
0134
0135 def main():
0136     validation = validate_and_restore(
0137         ROOT / "data/raw/german_credit_data.csv",
0138         ROOT / "data/raw/target_source/german_credit_with_risk.csv",
0139         ROOT / "data/processed/german_credit_with_verified_target.csv",
0140         ROOT / "reports/data_provenance_and_target_validation.md",
0141     )
0142     df = pd.read_csv(ROOT / "data/processed/german_credit_with_verified_target.csv")
0143     outliers = eda_figures(df)
0144     X, y = df.drop(columns="credit_risk", df["credit_risk"])
0145     X_train, X_test, y_train, y_test = train_test_split(
0146         X, y, test_size=.20, random_state=RANDOM_STATE, stratify=y
0147     )
0148     cv = StratifiedKFold(n_splits=5, shuffle=True, random_state=RANDOM_STATE)
0149     searches = {
0150         "Tuned Logistic Regression": (
0151             LogisticRegression(max_iter=3000, random_state=RANDOM_STATE),
0152             {"model__C": [.01, .1, 1, 10, 100], "model__penalty": ["l1", "l2"], "model__solver": ["liblinear"]},
0153         ),
0154         "Tuned SVM": (
0155             SVC(probability=True, random_state=RANDOM_STATE),
0156             {"model__C": [.1, 1, 10], "model__kernel": ["linear", "rbf"], "model__gamma": ["scale", "auto", .01,
0157             .1], "model__class_weight": [None, "balanced"]},
0158         ),
0159         "Tuned Gaussian Naive Bayes": (
0160             GaussianNB(), {"model__var_smoothing": np.logspace(-12, -6, 7)}
0161         ),
0162     }
0163     metrics, fitted, tuning = [], {}, {}
0164     for name, (model, grid) in searches.items():
0165         # Only the training schema is supplied here; all learned preprocessing
0166         # statistics are fitted inside GridSearchCV's training folds.
0167         pipeline = Pipeline([("preprocess", build_preprocessor(X_train)), ("model", model)])
0168         search = GridSearchCV(pipeline, grid, cv=cv, scoring="roc_auc", n_jobs=-1, return_train_score=False)
0169         search.fit(X_train, y_train)
0170         prediction = search.predict(X_test)
0171         probability = search.predict_proba(X_test)[:, 1]
0172         metrics.append(metric_row(name, y_test, prediction, probability))
0173         fitted[name] = search.best_estimator_
0174         tuning[name] = {"best_params": search.best_params_, "best_cv_roc_auc": float(search.best_score_)}
0175
0176     scratch_preprocessor = build_preprocessor(X_train)
0177     X_train_t = scratch_preprocessor.fit_transform(X_train, y_train)
0178     X_test_t = scratch_preprocessor.transform(X_test)
0179     scratch = LogisticRegressionScratch().fit(X_train_t, y_train.to_numpy())
0180     scratch_pred = scratch.predict(X_test_t)
0181     scratch_prob = scratch.predict_proba(X_test_t)[:, 1]
0182     metrics.append(metric_row("Logistic Regression from Scratch", y_test, scratch_pred, scratch_prob))
0183     plt.plot(scratch.loss_history_, color="#35618D")
0184     plt.title("Scratch logistic-regression convergence"); plt.xlabel("Iteration"); plt.ylabel("Binary cross-entropy + L2")
0185     save("11_logistic_regression_scratch_convergence.png")
0186
0187     comparison = pd.DataFrame(metrics).sort_values("ROC_AUC", ascending=False).reset_index(drop=True)
0188     comparison.to_csv(TABLE / "model_comparison.csv", index=False)
0189     header = "| Model | Accuracy | Precision | Recall | F1 | ROC-AUC |\n|---|---|---|---|---|---|\n"
0190     rows = "\n".join(
0191         f"| {r.Model} | {r.Accuracy:.3f} | {r.Precision:.3f} | {r.Recall:.3f} | {r.F1:.3f} | {r.ROC_AUC:.3f} |"
0192         for r in comparison.itertuples()
0193     )
0194     (TABLE / "model_comparison.md").write_text(header + rows + "\n", encoding="utf-8")
0195     pd.DataFrame([
0196         {"Model": k, "Best CV ROC-AUC": v["best_cv_roc_auc"], "Best parameters": json.dumps(v["best_params"])}
0197         for k, v in tuning.items()
0198     ]).to_csv(TABLE / "hyperparameter_tuning.csv", index=False)
0199     comparison.set_index("Model")[["Accuracy", "Precision", "Recall", "F1", "ROC_AUC']].plot.bar(figsize=(12,

```

```

6))
0200 plt.title("Test-set model comparison"); plt.ylabel("Score"); plt.ylim(0, 1); plt.legend(ncol=5, loc="lower
center", bbox_to_anchor=(.5, -0.32))
0201 save("12_model_metric_comparison.png")
0202 fig, ax = plt.subplots(figsize=(8, 6))
0203 for name, model in fitted.items():
0204     RocCurveDisplay.from_estimator(model, X_test, y_test, name=name, ax=ax)
0205 RocCurveDisplay.from_predictions(y_test, scratch_prob, name="Scratch Logistic", ax=ax)
0206 ax.plot([0, 1], [0, 1], "--", color="grey"); ax.set_title("Test-set ROC curves")
0207 save("13_roc_curves.png")
0208 fig, axes = plt.subplots(2, 2, figsize=(11, 9))
0209 all_predictions = {name: model.predict(X_test) for name, model in fitted.items()}
0210 all_predictions["Scratch Logistic"] = scratch_pred
0211 for ax, (name, prediction) in zip(axes.flat, all_predictions.items()):
0212     ConfusionMatrixDisplay(confusion_matrix(y_test, prediction), display_labels=["Good", "Bad"]).plot(ax=ax,
colorbar=False)
0213     ax.set_title(name)
0214 save("14_confusion_matrices.png")
0215
0216 logistic = fitted["Tuned Logistic Regression"]
0217 names = logistic.named_steps["preprocess"].get_feature_names_out()
0218 coefficients = logistic.named_steps["model"].coef_[0]
0219 coef = pd.DataFrame({"Feature": names, "Coefficient": coefficients})
0220 coef["Absolute coefficient"] = coef["Coefficient"].abs()
0221 coef = coef.sort_values("Absolute coefficient", ascending=False)
0222 coef.to_csv(TABLE / "logistic_feature_coefficients.csv", index=False)
0223 top = coef.head(15).sort_values("Coefficient")
0224 top.plot.barh(x="Feature", y="Coefficient", legend=False, color=["#35618D" if x < 0 else "#B64B4B" for x in
top["Coefficient"]])
0225 plt.title("Most influential logistic-regression coefficients"); plt.xlabel("Coefficient (positive = higher
bad-credit log-odds)"); plt.ylabel("")
0226 save("15_feature_importance.png")
0227
0228 best_name = comparison.iloc[0]["Model"]
0229 selected_name = "Tuned Logistic Regression" if comparison.loc[comparison.Model == "Tuned Logistic
Regression", "ROC_AUC"].iloc[0] >= comparison.ROC_AUC.max() - .03 else best_name
0230 selected = fitted.get(selected_name, fitted[best_name])
0231 fairness = []
0232 selected_pred = selected.predict(X_test)
0233 for group in sorted(X_test["Sex"].unique()):
0234     mask = X_test["Sex"].eq(group).to_numpy()
0235     fairness.append({"Sex": group, "n": int(mask.sum()), "Accuracy": accuracy_score(y_test.to_numpy()[mask],
selected_pred[mask]), "Recall_bad": recall_score(y_test.to_numpy()[mask], selected_pred[mask],
zero_division=0)})
0236 pd.DataFrame(fairness).to_csv(TABLE / "fairness_by_sex.csv", index=False)
0237 reports = {name: classification_report(y_test, pred, output_dict=True) for name, pred in
all_predictions.items()}
0238 results = {
0239     "validation": validation, "shape": list(df.shape), "missing_counts": X.isna().sum().to_dict(),
0240     "outlier_counts_qqr": outliers, "train_rows": len(X_train), "test_rows": len(X_test),
0241     "split_strategy": {
0242         "test_size": 0.20, "random_state": RANDOM_STATE, "stratified": True,
0243         "training_class_counts": {str(k): int(v) for k, v in y_train.value_counts().sort_index().items()},
0244         "testing_class_counts": {str(k): int(v) for k, v in y_test.value_counts().sort_index().items()},
0245         "training_class_percent": {str(k): float(v * 100) for k, v in
y_train.value_counts(normalize=True).sort_index().items()},
0246         "testing_class_percent": {str(k): float(v * 100) for k, v in
y_test.value_counts(normalize=True).sort_index().items()},
0247         "cross_validation": "StratifiedKFold(n_splits=5, shuffle=True, random_state=42)",
0248         "grid_search_input": "X_train and y_train only",
0249         "threshold_optimisation": "None; fixed decision threshold 0.5",
0250     },
0251     "tuning": tuning, "metrics": comparison.to_dict(orient="records"), "selected_model": selected_name,
0252     "selection_rule": "Prefer the tuned interpretable logistic model when within 0.03 ROC-AUC of the highest
model; otherwise select the highest ROC-AUC model.",
0253     "scratch": {"iterations": scratch.n_iter_, "converged": scratch.converged_, "final_loss":
scratch.loss_history_[-1]},
0254     "top_coefficients": coef.head(10).to_dict(orient="records"), "fairness_by_sex": fairness,
0255     "classification_reports": reports,
0256     "environment": {"python": sys.version.split()[0], "platform": platform.platform(), "pandas":
pd.__version__, "numpy": np.__version__, "scikit_learn": sklearn.__version__},
0257 }
0258 (TABLE / "pipeline_results.json").write_text(json.dumps(results, indent=2), encoding="utf-8")
0259 evidence_card("Dataset loading and validation", [
0260     "Rows: 1,000 | Predictors: 9 | Target: credit_risk",
0261     "Row-level predictor reconciliation: PASS (100%)",
0262     "Mismatched predictor cells: 0",
0263     "Class balance: 700 good / 300 bad",
0264     "Missing: Saving accounts=183; Checking account=394",
0265 ], "01_dataset_validation.png")
0266 evidence_card("Model execution evidence", [
0267     *["{r.Model}: AUC={r.ROC_AUC:.3f}, Recall={r.Recall:.3f}, F1={r.F1:.3f}" for r in
comparison.itertuples()],
0268     f"Selected model: {selected_name}",
0269     f"Scratch convergence: {scratch.converged_}; iterations={scratch.n_iter_}",
0270 ], "02_model_results.png")
0271 academic_summary_card("Assessment Workflow Status", [
0272     "Dataset successfully validated",
0273     "Target labels verified and aligned",
0274     "Data preprocessing completed",
0275     "Training/test split completed (800/200)",
0276     "10 EDA figures and 5 evaluation figures generated",
0277     "No data leakage detected",
0278 ], "03_pipeline_completion.png")
0279 print(json.dumps({"selected_model": selected_name, "metrics": results["metrics"], "validation": "PASS"},
indent=2))
0280
0281
0282 if __name__ == "__main__":
0283     main()

```


scripts/validate_submission.py

```
0001 """Fail-fast validation for the single-attempt LMS submission."""
0002 from __future__ import annotations
0003
0004 import json
0005 from pathlib import Path
0006
0007 import pdfplumber
0008
0009 ROOT = Path(__file__).resolve().parents[1]
0010 CORRECT_EMAIL = "b.emeokoro6653@miva.edu.ng"
0011 DISALLOWED_EMAIL = CORRECT_EMAIL.rsplit(".", 1)[0] + ".org"
0012
0013
0014 def main():
0015     required = [
0016         "README.md", "notebooks/MIT8301_CA_German_Credit_Risk.ipynb",
0017         "reports/MIT8301_CA_Report.md", "reports/MIT8301_CA_Report.pdf",
0018         "reports/data_provenance_and_target_validation.md", "reports/tables/model_comparison.csv",
0019         "submission/Basil_Oforbuike_Emeokoro_MIT8301_CA.pdf",
0020     ]
0021     missing = [name for name in required if not (ROOT / name).exists()]
0022     assert not missing, f"Missing required files: {missing}"
0023     results = json.loads((ROOT / "reports/tables/pipeline_results.json").read_text(encoding="utf-8"))
0024     assert results["validation"]["validation_passed"] and results["validation"]["total_mismatched_cells"] == 0
0025     assert len(results["metrics"]) == 4 and len(list((ROOT / "reports/figures").glob("*.png"))) >= 15
0026     notebook = json.loads((ROOT / "notebooks/MIT8301_CA_German_Credit_Risk.ipynb").read_text(encoding="utf-8"))
0027     assert all(cell.get("execution_count") is not None for cell in notebook["cells"] if cell["cell_type"] ==
0028               "code")
0029     text_files = list(ROOT.rglob("*.md")) + list(ROOT.rglob("*.py")) + list(ROOT.rglob("*.ipynb"))
0030     combined = "\n".join(path.read_text(encoding="utf-8") for path in text_files)
0031     assert DISALLOWED_EMAIL not in combined
0032     assert CORRECT_EMAIL in combined
0033     final_pdf = ROOT / "submission/Basil_Oforbuike_Emeokoro_MIT8301_CA.pdf"
0034     with pdfplumber.open(final_pdf) as pdf:
0035         page_count = len(pdf.pages)
0036         assert page_count >= 20
0037         first_text = "\n".join((page.extract_text() or "") for page in pdf.pages[:3])
0038         assert CORRECT_EMAIL in first_text
0039     checklist = f"""# Final Submission Checklist
0040 - [x] Target-source reconciliation passed with zero mismatched cells.
0041 - [x] Verified 1,000 rows and 700/300 class balance.
0042 - [x] Notebook contains executed outputs.
0043 - [x] Accuracy, precision, recall, F1, and ROC-AUC are complete for four models.
0044 - [x] At least 15 actual-output figures exist.
0045 - [x] Report and final single-file PDF exist and are non-empty.
0046 - [x] Student details use {CORRECT_EMAIL} consistently.
0047 - [x] All 40 rubric marks are mapped in the report.
0048 - [x] Unit and smoke tests pass.
0049 - [x] No credentials or machine-specific paths appear in user instructions.
0050 """
0051     (ROOT / "submission/submission_checklist.md").write_text(checklist, encoding="utf-8")
0052     print(f"PASS: {len(required)} required artifacts, {len(list((ROOT / 'reports/figures').glob('*.png')))}
0053           figures, {page_count} PDF pages")
0054
0055 if __name__ == "__main__":
0056     main()
```


scripts/validate_target_source.py

```
0001 from run_pipeline import main
0002
0003 if __name__ == "__main__":
0004     main()
```

tests/test_data_validation.py

```
0001 from pathlib import Path
0002 import pandas as pd
0003
0004 from mit8301_credit_risk.data_validation import validate_and_restore
0005
0006
0007 ROOT = Path(__file__).resolve().parents[1]
0008
0009
0010 def test_exact_target_reconciliation(tmp_path):
0011     output = tmp_path / "restored.csv"
0012     report = tmp_path / "report.md"
0013     result = validate_and_restore(ROOT / "data/raw/german_credit_data.csv", ROOT /
0014     "data/raw/target_source/german_credit_with_risk.csv", output, report)
0015     restored = pd.read_csv(output)
0016     assert result["validation_passed"]
0017     assert result["total_mismatched_cells"] == 0
0018     assert restored.shape == (1000, 10)
0019     assert restored.credit_risk.value_counts().to_dict() == {0: 700, 1: 300}
0020
0021 def test_mismatch_stops_reconciliation(tmp_path):
0022     supplied = pd.DataFrame({"Age": [20], "Sex": ["male"]})
0023     candidate = pd.DataFrame({"Age": [21], "Sex": ["male"], "Risk": ["good"]})
0024     supplied.to_csv(tmp_path / "a.csv", index=False); candidate.to_csv(tmp_path / "b.csv", index=False)
0025     try:
0026         validate_and_restore(tmp_path / "a.csv", tmp_path / "b.csv", tmp_path / "o.csv", tmp_path / "r.md")
0027     except ValueError as exc:
0028         assert "mismatched" in str(exc)
0029     else:
0030         raise AssertionError("A predictor mismatch must stop target restoration")
```

tests/test_logistic_regression_scratch.py

```
0001 import numpy as np
0002 from sklearn.datasets import make_classification
0003 from sklearn.linear_model import LogisticRegression
0004 from sklearn.metrics import accuracy_score
0005 from sklearn.preprocessing import StandardScaler
0006
0007 from mit8301_credit_risk.logistic_regression_scratch import LogisticRegressionScratch
0008
0009
0010 def test_sigmoid_and_probability_bounds():
0011     values = LogisticRegressionScratch._sigmoid(np.array([-1000, 0, 1000]))
0012     assert np.all((values >= 0) & (values <= 1))
0013     assert values[1] == 0.5
0014
0015
0016 def test_loss_decreases_and_predictions_agree_reasonably():
0017     X, y = make_classification(n_samples=500, n_features=8, random_state=42)
0018     X = StandardScaler().fit_transform(X)
0019     model = LogisticRegressionScratch(max_iter=3000).fit(X, y)
0020     benchmark = LogisticRegression(max_iter=2000).fit(X, y)
0021     assert model.loss_history_[-1] < model.loss_history_[0]
0022     assert set(model.predict(X)) <= {0, 1}
0023     assert np.all((model.predict_proba(X) >= 0) & (model.predict_proba(X) <= 1))
0024     assert abs(accuracy_score(y, model.predict(X)) - accuracy_score(y, benchmark.predict(X))) < 0.08
```

tests/test_pipeline_smoke.py

```
0001 import json
0002 from pathlib import Path
0003
0004 ROOT = Path(__file__).resolve().parents[1]
0005
0006
0007 def test_generated_pipeline_outputs_are_complete():
0008     results = json.loads((ROOT / "reports/tables/pipeline_results.json").read_text(encoding="utf-8"))
0009     assert results["validation"]["validation_passed"]
0010     assert len(results["metrics"]) == 4
0011     for model in results["metrics"]:
0012         assert {"Accuracy", "Precision", "Recall", "F1", "ROC_AUC"} <= model.keys()
0013     assert (ROOT / "submission/Basil_Oforbuike_Emeokoro_MIT8301_CA.pdf").stat().st_size > 100_000
```

tests/test_preprocessing.py

```
0001 from pathlib import Path
0002 import pandas as pd
0003 from sklearn.model_selection import train_test_split
0004
0005 from mit8301_credit_risk.preprocessing import build_preprocessor
0006
0007
0008 ROOT = Path(__file__).resolve().parents[1]
0009
0010
0011 def test_preprocessor_fits_training_data_and_handles_unknowns():
0012     df = pd.read_csv(ROOT / "data/processed/german_credit_with_verified_target.csv")
0013     X_train, X_test = train_test_split(df.drop(columns="credit_risk"), test_size=.2, random_state=42)
0014     processor = build_preprocessor(X_train).fit(X_train)
0015     transformed_train = processor.transform(X_train)
0016     changed = X_test.copy(); changed.loc[changed.index[0], "Purpose"] = "unseen-purpose"
0017     transformed_test = processor.transform(changed)
0018     assert transformed_train.shape[1] == transformed_test.shape[1]
0019     assert hasattr(processor.named_transformers_["num"].named_steps["clipper"], "lower_")
```

tests/test_train_test_strategy.py

```
0001 from pathlib import Path
0002
0003 import pandas as pd
0004 from sklearn.model_selection import train_test_split
0005
0006
0007 ROOT = Path(__file__).resolve().parents[1]
0008
0009
0010 def test_stratified_split_counts_and_percentages():
0011     data = pd.read_csv(ROOT / "data/processed/german_credit_with_verified_target.csv")
0012     X = data.drop(columns="credit_risk")
0013     y = data["credit_risk"]
0014     _, y_train, y_test = train_test_split(
0015         X, y, test_size=0.20, random_state=42, stratify=y
0016     )
0017     assert y_train.value_counts().sort_index().to_dict() == {0: 560, 1: 240}
0018     assert y_test.value_counts().sort_index().to_dict() == {0: 140, 1: 60}
0019     assert (y_train.value_counts(normalize=True).sort_index() * 100).to_dict() == {0: 70.0, 1: 30.0}
0020     assert (y_test.value_counts(normalize=True).sort_index() * 100).to_dict() == {0: 70.0, 1: 30.0}
0021
0022
0023 def test_source_keeps_tuning_and_preprocessing_off_test_set():
0024     source = (ROOT / "scripts/run_pipeline.py").read_text(encoding="utf-8")
0025     assert "test_size=.20, random_state=RANDOM_STATE, stratify=y" in source
0026     assert "search.fit(X_train, y_train)" in source
0027     assert "search.fit(X, y)" not in source
0028     assert "build_preprocessor(X_train)" in source
0029     assert "fit_transform(X_train, y_train)" in source
0030     assert "transform(X_test)" in source
0031     assert "StratifiedKFold(n_splits=5, shuffle=True, random_state=RANDOM_STATE)" in source
```